

Structure of fungal tRNA ligase Trl1 with RNA reveals conserved substrate-binding principles

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RNA ligases play a vital role in RNA processing and maturation, including tRNA splicing, RNA repair and the unfolded protein response (UPR). In fungi and plants, the tripartite tRNA ligase Trl1 catalyzes the joining of TSEN-cleaved pre-tRNA exon halves. Trl1 also functions as ligase in the non-conventional *HAC1* mRNA splicing during the UPR. The final ligation step is performed by the N-terminal adenylyltransferase domain (ligase; LIG). The spatial arrangement of the exon ends during the ligation reaction has remained elusive. Here we report the crystal structure of *Chaetomium thermophilum* Trl1-LIG in complex with a tRNA-derived substrate. Our structure represents a snapshot of the activated RNA intermediate and defines the conserved substrate-binding interface. The underlying enzyme-substrate interplay reveals a substrate-binding principle shared by adenylyltransferases. Moreover, we identify the determinants of RNA end specificity as well as the specific roles of Trl1-LIG's subdomains during ligase activation, substrate binding and phosphoryl transfer.

The ligation of RNA ends by RNA ligases is crucial for RNA maturation, processing and repair in all domains of life. Various RNA ligases have evolved to act on diverse RNA substrates¹. Non-spliceosomal transfer RNA (tRNA) splicing is an essential step in tRNA biosynthesis². In eukaryotes, certain pre-tRNAs contain an intronic sequence, located one base 3' from the anticodon³. These introns disrupt the RNA tertiary structure of the anticodon stem-loop (ASL). During the enzyme-catalyzed splicing reaction, a tRNA-splicing endonuclease (SEN in yeast; TSEN in vertebrates) first excises the intron, resulting in two exon halves that are subsequently connected by a tRNA ligase^{4–6}. After cleavage by the SEN or TSEN complex, the exon halves contain a 2',3'-cyclic phosphate (cP) at the 5' exon and a 5'-OH group at the splice site of the 3' exon^{6–8}. Despite conceptual similarities, the subsequent enzymatic reaction of tRNA exon–exon joining varies by organism⁹. In eukaryotes, two main mechanisms exist: one in fungi and plants, and one in metazoans (including humans). In fungi, the tRNA exon halves are ligated by the tRNA ligase Trl1 (previously called Rlg1)^{5,10,11}; in metazoans, they

are joined by RTCB as part of a multi-protein ligase complex⁸. Both ligases also catalyze the non-conventional splicing of *HAC1* and *XBPI* mRNA during the unfolded protein response^{12–15}. In fungi, chemical modification at both RNA termini precedes the ligation of the pre-tRNA exon halves (Fig. 1a)⁵. The tripartite enzyme Trl1 catalyzes all steps and consists of (1) an amino-terminal adenylyltransferase domain (LIG), which belongs to the nucleotidyltransferase superfamily including DNA ligases and RNA-capping enzymes, (2) a carboxy-terminal cyclic phosphodiesterase domain (CPD) and (3) a central polynucleotide kinase (KIN) domain^{11,16}. The CPD domain opens the 2',3'-cP to form a 3'-OH and 2'-phosphate (P), and the KIN activity phosphorylates the 5'-OH in a GTP-dependent reaction. Both modifications prepare the RNA substrate for the ATP-dependent, three-step ligation by the LIG domain (Fig. 1a). First, the reaction between the active site lysine and ATP yields a covalent LIG–(lysyl- ζ)–AMP intermediate. Subsequently, the AMP is transferred to the 5'-P end of the 3' exon and results in a 5'-5' RNA-adenylylate (AppRNA). Finally, the LIG domain catalyzes

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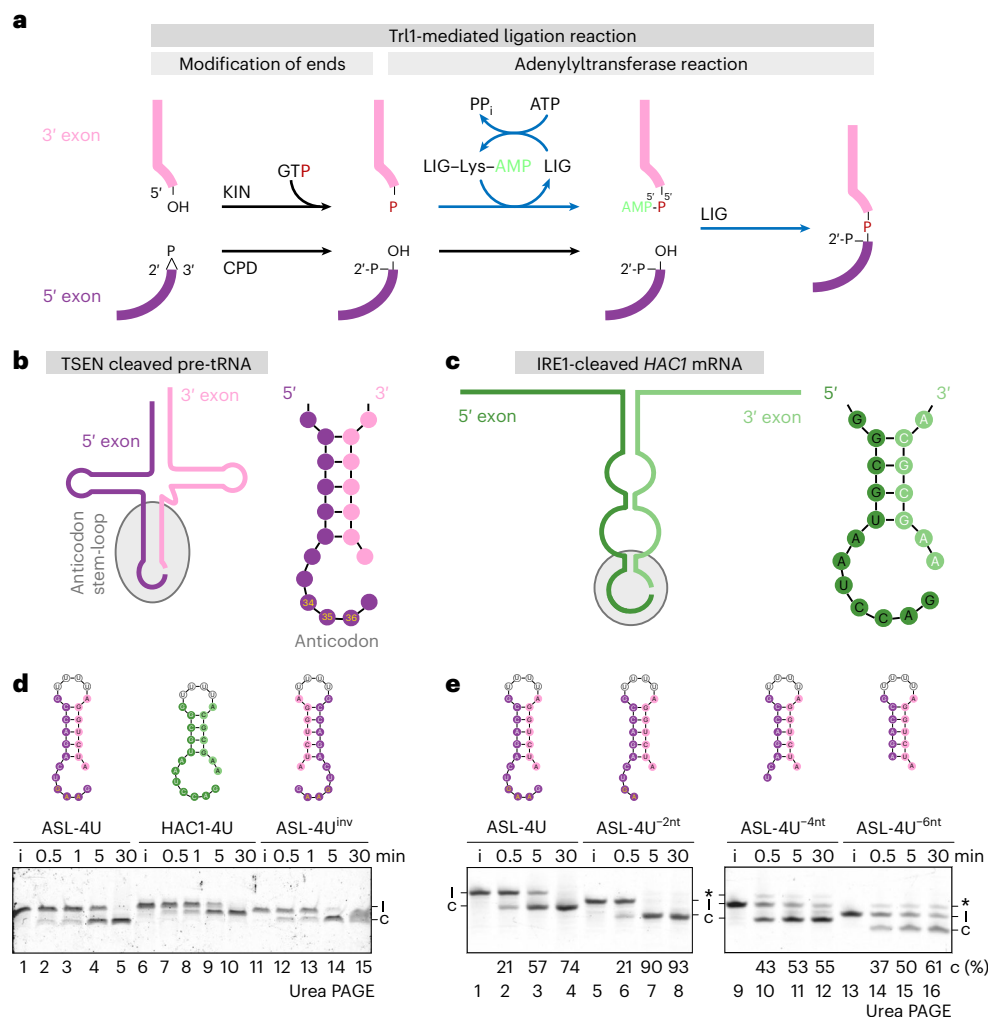


Fig. 1 | Trl1 ligation mechanism and RNA substrate structures. a, Overview of the tRNA ligation reaction by the fungal tRNA ligase Trl1. The RNA ends at the splice site of both exons (5' exon in purple and 3' exon in pink) are enlarged, with the respective chemical groups at the termini. During the Trl1-catalyzed reaction, the exon ends are first modified by the KIN and the CPD domain, followed by the adenylyltransferase reaction of the LIG domain, resulting in phosphodiester bond formation. The same ligation mechanism joins the non-conventional HAC1 mRNA exons during the unfolded protein response. **b**, The TSEN cleaved pre-tRNA halves are depicted as secondary structure, with the 5' exon (purple) and the 3' exon (pink). The ASL is highlighted with a gray border and enlarged on the right. The ASL folds in the same way for all cleaved pre-tRNAs: a 5-bp stem-loop with 6 unpaired nucleotides (including the anticodon triplet at position 34–36 in yellow) at the 5' splice site and 1 unpaired nucleotide at the 3' splice site. **c**, The IRE1-cleaved HAC1 mRNA is depicted as secondary structure with the 5' exon

(dark green) and the 3' exon (light green). The stem loop with the splice site is highlighted with a gray border and enlarged on the right. The structure consists of a 4-bp stem-loop with 7 unpaired nucleotides at the 5' splice site and 2 unpaired nucleotides at the 3' splice site. **d**, Circularization assay of CtTrl1-LIG WT (1 μM) with the linear ASL-4U, HAC1-4U and ASL-4U^{inv} substrates. The ligation of the linear RNA substrate (l) to the circular form (c) was monitored by urea PAGE over time, performed in triplicate (n = 3). **e**, Circularization assay of WT CtTrl1-LIG (0.1 μM) with varying length of the single-stranded overhangs at the 5' exon 3' end (ASL-4U^{-2nt}, ASL-4U^{-4nt} and ASL-4U^{-6nt}). Ligation of the linear ASL-4U substrates (l) to the circular form (c), as well as formation of the AppRNA intermediate (*), was monitored by urea PAGE. The percentage of circularization product was calculated as the ratio of circularized to non-circularized RNA from one duplicate (n = 2). i, input.

the nucleophilic attack by the 3'-OH on the AppRNA to form a phosphodiester bond, thereby releasing AMP^{5,11}. The remaining 2'-P at the splice junction is removed by the essential 2'-phosphodiesterase Tpt1 (refs. 17–20). The importance of a 2'-P for the adenylyltransferase reaction is a characteristic of Trl1-type RNA-ligases.

Crystal structures of the LIG domain from the thermophilic fungus *Chaetomium thermophilum* (syn. *Thermochaetoides thermophila*) have been solved^{21,22}. The LIG domain features a bi-lobal architecture, comprising an N-terminal canonical adenylyltransferase domain (NTD) and a C-terminal all-helical domain (CTD). This design, combining an adenylyltransferase core with a unique CTD, also exists in other RNA and DNA ligases^{23–25}. Studies on the bacteriophage T4 tRNA-repair ligase Rnl1 show that its CTD confers specificity for tRNA substrates²⁴. The structure of the RNA-editing ligase KREL1 from *Trypanosoma brucei*

revealed a diverged CTD with an unknown function²⁵. The specific role of the CTD within Trl1-type RNA ligases has remained elusive.

Crystal structures of ATP-dependent RNA ligases have revealed conformations associated with different nucleotide-bound states^{24,26–29}. However, the existing structural information on substrate binding and coordination of the exon ends is scarce. The crystal structure of the nick-sealing RNA ligase Rnl2 from bacteriophage T4 in complex with a nicked DNA duplex provides insights into the substrate-binding mechanisms of dsRNA ligases³⁰. In addition, structures of DNA ligases with bound substrates have revealed the binding modes of adenylyltransferases to nucleic acids^{31–33}. However, our understanding of how RNA ligases coordinate both exon ends when acting on single-stranded RNA substrates remains limited. Furthermore, the structural basis for the 2'-P specificity of Trl1-type ligases is unknown. Here, we present the

crystal structure of *C. thermophilum* Trl-LIG (*CtTrl1*-LIG) in complex with an activated RNA substrate, revealing insights into substrate binding, activation and specificity. Complementary biochemical analyses dissect the role of the CTD and identify the determinants of 2'-P requirement. On the basis of molecular simulations, we propose a model for binding of physiological tRNA substrates by Trl1.

Trl1 substrates consist of an RNA duplex stem with overhangs

To understand the principles underlying LIG-RNA interaction and exon-exon coordination, we compiled the predicted RNA structures of both physiological substrates: TSEN-cleaved pre-tRNAs and Ire1-cleaved *HAC1* mRNA (Fig. 1b,c and Extended Data Fig. 1a).

Recent structures of the TSEN complex with pre-tRNA substrates reveal that cleaved pre-tRNAs and mature tRNAs assume the same fold^{34–36}. There are six unpaired nucleotides at the 5' splice site and one unpaired nucleotide at the 3' splice site (Fig. 1b). Given that the intron position is mostly invariable in eukaryotes³⁷, all cleaved pre-tRNAs in *Saccharomyces cerevisiae* consist of a stem of 5 base pairs (bp) with single-stranded overhangs (Fig. 1b and Extended Data Fig. 1a). For *HAC1* mRNA, base-pairing keeps both exons together after cleavage by Ire1 (ref. 38). In the predicted *HAC1* mRNA structure, the ends at the splice site extend as overhangs from a duplex stem, with seven unpaired nucleotides at the 5' splice site and two unpaired nucleotides at the 3' splice site (Fig. 1c).

To assess the requirements of an RNA substrate and the effect of RNA structure on Trl1-LIG-catalyzed ligation, we tested different substrates derived from pre-tRNA anticodon stem-loops and the Ire1-cleaved *HAC1* mRNA stem-loop (Fig. 1b,c). We generated a bona fide tRNA ASL substrate derived from yeast tRNA^{Phe} by linking both exon strands with four uridines (4U) (Extended Data Fig. 1b; see Supplementary Table 2 for details). Likewise, a *HAC1*-stem loop 4U-linked substrate (*HAC1*-4U) was designed. Ligation of either ASL-4U or *HAC1*-4U by *CtTrl1*-LIG resulted in circularized ligation products with similar kinetics (Fig. 1d, lanes 1–10). In both physiological substrates, the single-stranded overhang at the 5' splice site is longer than the 3' splice site. To test whether the length of the overhangs affects ligation, we performed circularization using a substrate with inverted 5' and 3' exons (ASL-4U^{inv}). *CtTrl1*-LIG ligated ASL-4U^{inv} to completion, with similar kinetics to those of the bona fide substrate (Fig. 1d, lanes 11–15). Inversion of the 5' and 3' overhangs in ASL-4U also resulted in ligation (Extended Data Fig. 1c).

We further tested whether the length of the overhang at the 5' splice site impacts ligation kinetics by comparing ASL-4U with the canonical 6-nucleotide (nt) overhang with substrates shortened by 2 nt, 4 nt and 6 nt (Fig. 1e). Although we observed complete circularization for ASL-4U and ASL-4U^{2nt}, a considerable fraction remained unligated for ASL-4U^{4nt} and ASL-4U^{6nt}. Despite incomplete ligation, early timepoints showed more circularized product (Fig. 1e, compare lane 2 with lanes 10 and 14). We speculate that the –4-nt and –6-nt RNA substrates formed two distinct pools, a ligation-incompetent one and a ligation-competent one, which surpassed ligation kinetics of the canonical substrate. Such separation could explain the complex (presumably biphasic) kinetics. Moreover, we detected accumulation of the AppRNA intermediate (Fig. 1e, lanes 10–12 and 14–16, marked by *), indicating disturbed RNA end coordination of the truncated strand in the LIG active site.

Additionally, we examined whether LIG prefers a canonical ASL over an entirely single-stranded RNA (composed of A and G only) of the same length. Circularization of the tRNA-derived ASL-4U RNA was faster than that of unstructured RNA (Extended Data Fig. 1d, lanes 1–10). We noticed that the circularized ligation product of the structured RNA substrate migrated faster than the single-stranded A-G-only substrate despite denaturing conditions. Thus, we used an equivalent, folded stem-loop RNA with only A-U base pairs in the

stem area, which confirmed the difference in migration behavior and faster ligation kinetics for stem-loop substrates (Extended Data Fig. 1d, lanes 11–15).

Using electrophoretic mobility shift assays (EMSAs), we uncovered a slightly decreased affinity to the RNA without overhang (ASL-4U^{6nt}) compared with the canonical stem-loop substrate (Extended Data Fig. 1e).

Taken together, these analyses showed that a combination of RNA folding and overhang length at the splice sites determines the efficiency of exon-exon ligation.

Structure of *CtTrl1*-LIG with an activated RNA substrate

To gain structural insights into binding and coordination of RNA by the LIG domain, we aimed to crystallize *CtTrl1*-LIG with an RNA substrate. We screened various RNA substrates derived from both physiological Trl1 RNA substrates, cleaved pre-tRNA or *HAC1* mRNA. We focused on using shortened RNA substrates to minimize the conformational flexibility outside of Trl1-LIG's RNA-binding cleft. On the basis of RNA secondary structure prediction³⁹ and substrate requirements identified in biochemical analyses (Fig. 1 and Extended Data Fig. 1), we attempted to crystallize *CtTrl1*-LIG with duplex-forming RNA oligonucleotides with short single-stranded overhangs in the presence of ATP. We obtained diffracting microcrystals in the space group *P*2₁2₁2₁ for *CtTrl1*-LIG (residues 1–414), in complex with the first 7 nt of the 3' exon from cleaved yeast pre-tRNA^{Phe}, featuring a 5'-P end (5'-P-AUCUGGA-3'). This RNA substrate was predicted to form a duplex owing to its self-complementarity. We determined the LIG-RNA complex structure at 2.37-Å resolution using molecular replacement (Fig. 2a and Table 1). Indeed, the RNA substrate crystallized as a bound duplex (Fig. 2a). Our structure contained AMP in the LIG active site in the same orientation as in previously solved structures. We observed continuous electron density between the α -phosphate of AMP and the 5'-P of one of the RNA strands, indicative of the formation of an activated AppRNA intermediate (Fig. 2b). The AMP moiety of the activated AppRNA intermediate is coordinated by residues T146, K148 and K325 (Extended Data Fig. 2a). The continuing 3' exon is further contacted by R99, K169 and K323. All of these residues are located in the NTD. Although the assembly of the ligase-RNA complex in the crystallographic unit cell promoted activation of the 5' end, it did not position the 3' end of the complementary strand correctly for ligation, and we did not observe any protein-RNA interactions with this strand (see Fig. 2a, sand-colored). We therefore attribute no physiological relevance to this strand, which instead served in stabilizing the activated strand to enable crystallization.

To position the obtained LIG-RNA structure in the three-step ligation mechanism, we compared with previously reported structures of *CtTrl1*-LIG in the ATP- and AMPcPP-bound state, as well as the activated LIG-Lys-AMP state (Extended Data Fig. 2b)^{21,22}. Overall, the conformational changes between the three states are subtle. Superposition with adenylylated LIG (PDB ID 6NOV) shows that the adenine base, ribose and α P of the AMP cofactor remain in fixed positions during adenylyl transfer (Extended Data Fig. 2c). The orientation of the broken and newly formed bonds at the α P support in-line attack. The phosphoryl transfer coincides with a stereochemical inversion at the α P center. The lysine at the catalytic site (K148 in *CtTrl1*-LIG) shows minimal movement between both states. We conclude that our LIG-RNA structure represents a state immediately following the adenylyl transfer from the active site lysine onto the 5' RNA end (that is post step 2, see Extended Data Fig. 2d). The Mn²⁺ ion, which is present in the LIG-AMP structure, is absent in the LIG-RNA structure.

Notably, the cocrystallized RNA duplex did not demonstrate a 1:1 interaction with LIG. Instead, in the crystal lattice of the substrate complex, the RNA duplexes occupy the space between neighboring LIG symmetry mates (Fig. 2c,d) by bridging two protein molecules (Fig. 2e). This crystal packing provides further insights into substrate

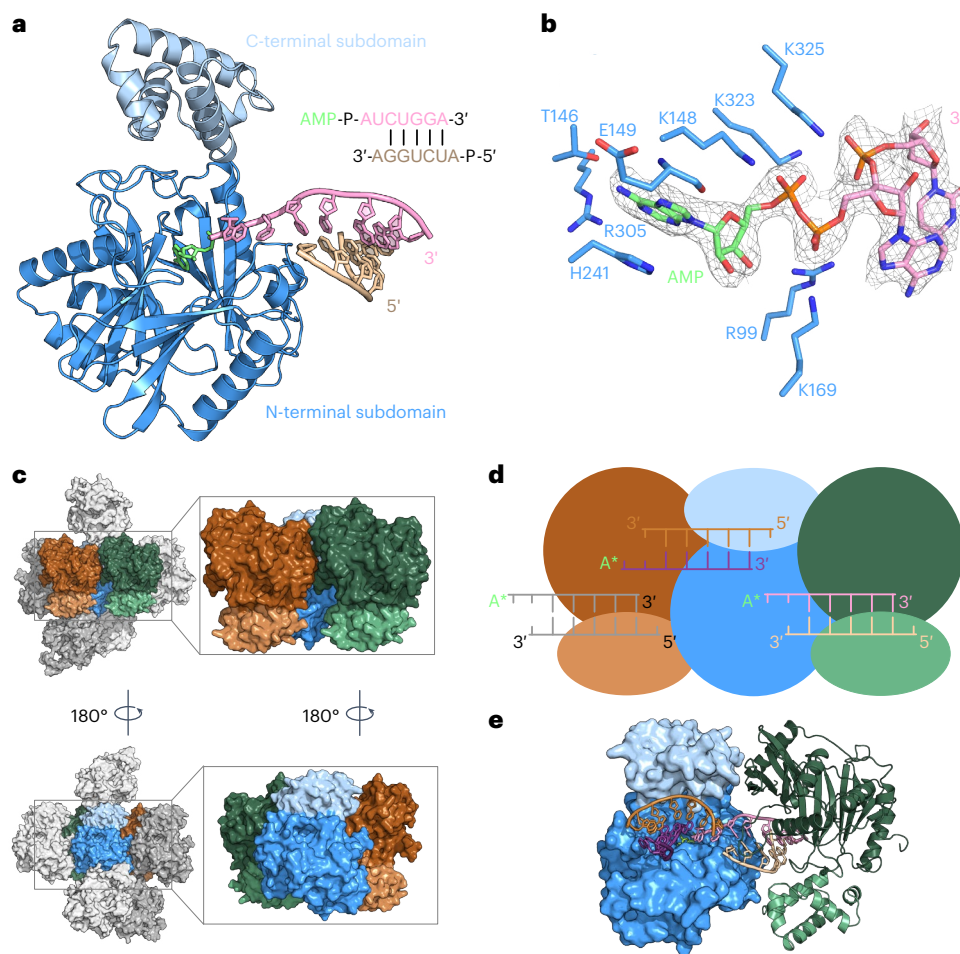


Fig. 2 | Structure of *CrTrlI*-LIG with an activated RNA substrate. a, Cartoon representation of the overall structure of *CrTrlI*-LIG in complex with an activated RNA substrate. The N-terminal domain is colored blue, and the C-terminal domain is light blue. The AMP cofactor (lime) is depicted as sticks and the RNA strands (pink and sand) are shown in cartoon style. The RNA sequence overview contains the corresponding colors. **b**, A zoomed-in view of the active site with the activated AppRNA 5' end, with the coordinating amino acid residues depicted as sticks. The $2F_o - F_c$ electron density OMIT map at 1σ is shown as a gray mesh. **c**, The nearest symmetry mates (relative to the blue molecule) of the LIG-RNA

complex in the $P2_12_12_1$ crystal. The individual molecules are shown in surface depiction. Two symmetry mates of the assembly (colored in orange and green) are enlarged for simplification (black rectangles). **d**, Scheme of the arrangement in **c**, illustrating the end-to-end bridging of LIG molecules and their active sites by the bound RNA substrate molecules (same color code as in **c**). **e**, Surface representation of *CrTrlI*-LIG (blue) with bound RNA duplexes (bronze and purple, pink and sand) as well as a neighboring symmetry mate in cartoon depiction (green) visualizing the binding of one RNA substrate between two LIG molecules in the crystal.

binding from the perspective of one LIG molecule and its RNA contacts within the crystal (Fig. 2e, blue LIG molecule).

An extended RNA-binding cleft determines RNA end coordination

When we analyzed all RNA-protein interactions in the crystal, we found that a single LIG molecule binds two RNA duplexes in its extended substrate binding cleft (Fig. 3a). In addition to the RNA duplex, defined in the asymmetric unit (ASU) (Fig. 2a), a second duplex occupies the opposite half of the binding cleft (Fig. 3a and Extended Data Fig. 3a). This binding mode suggests that there is a double-stranded RNA (dsRNA)-binding surface in LIG that utilizes both the NTD and CTD to cradle the RNA duplex (Fig. 3a, duplex on the left). The CTD contacts one strand of the duplex with only one residue, N382 (Fig. 3b; compare with the bronze-colored RNA strand in Fig. 3a). The other strand of the additional duplex is coordinated by a cluster of residues in the NTD, which localize to a loop segment spanning residues 168–181 (Fig. 3a, purple strand). Parts of this loop form a thumb-like protrusion at one end of the substrate-binding cleft (Extended Data Fig. 2d). This loop region exhibits the greatest conformational variability among the available *CrTrlI*-LIG structures (Extended Data Fig. 2b). Most RNA-binding

residues interact with the phosphate backbone (K64, Y68, R83, I153, S168, K169, H170, S171 and H182), and additional contacts (R175 and H182) to the 2'-OH groups allow differentiation between RNA and DNA (Fig. 3b). We did not observe any interactions with the RNA bases.

On the basis of the orientation of the RNA ends toward the active site, as well as LIG-RNA interactions, we interpret the ultimate 5' adenylated strand (Fig. 3b,c, pink RNA strand) as the 3'-exon equivalent and, consequently, the 4 nt of the strand from the other duplex as the 5'-exon equivalent (Fig. 3b,c, purple RNA strand). The 4.7-Å distance between the 3'-OH and the α P, as well as their relative orientation, suggest that additional repositioning of the end of the 5' exon is necessary for the final ligation step (Fig. 3d, zoom). The distance between the 5' and 3' ends must be closed and the 3'-OH must adopt a favorable orientation of approximately 180° toward the AppRNA phosphorus atom to facilitate in-line attack. We surmise that the presence of a 2'-P, which is a hallmark of TrlI substrates but absent in the crystal structure, would facilitate the required repositioning.

Conservation analysis^{40,41} revealed that the entire TrlI-LIG substrate-binding cleft, including all RNA-interacting residues in our LIG-RNA structure, is evolutionarily conserved (Extended Data Fig. 4). To determine the in vivo relevance of the RNA-binding residues, we

Table 1 | Crystallographic data collection and refinement

CrTrl1-LIG-RNA	
PDB ID	8RBJ
Data collection	
Space group	$P2_12_1$
Resolution (Å)	2.37–47.6 (2.37–2.46)
Unique reflection	20384 (2019)
Cell dimensions	
a, b, c (Å)	51.56, 75.36, 124.56
α, β, γ (°)	90, 90, 90
R_{merge}	0.355 (3.123)
R_{pim}	0.090 (0.856)
Mean $I/\sigma(I)$	7.7 (1.0)
Multiplicity	15.8 (13.6)
Completeness (%)	100 (100)
$CC_{1/2}$	0.989 (0.392)
Refinement	
R_{work} (%)	17.87 (29.59)
R_{free} (%)	21.78 (37.04)
r.m.s.d. bond length (Å)	0.007
r.m.s.d. bond angle (°)	0.950
Ramachandran favored (%)	97.4
Ramachandran allowed (%)	2.3
Ramachandran outliers (%)	0.3
Rotamer outliers (%)	1.2
Clash score	2.5
Average B factor (Å ²)	60.3
Protein	60.4
Ligands	48.2
Solvent	47.7

Values in parentheses refer to the highest-resolution shell. The R_{free} set consists of 5% randomly chosen data excluded from refinement. r.m.s.d., root mean square deviation.

used yeast-plasmid-based mutagenesis. A previous study in *S. cerevisiae* (3c) tested the viability of alanine substitutions for many residues that interact with RNA in our structure⁴². We substituted the remaining homologous residues with alanine and tested yeast growth (Extended Data Fig. 5). With the exception of I153, R175 and N382, the identified RNA-binding residues are essential for viability (Fig. 3b, bold labels). On the basis of our structural data and the composition of a cleaved pre-tRNA ASL (Fig. 1b), we schematically depicted the LIG–ASL complex (Fig. 3e).

Confirmation of the proposed binding surface at the 5' exon

The adenylation of the 5' end supports correct positioning of the 3'-exon-equivalent strand in our crystal structure. To test the proposed RNA-binding interface at the 5'-exon end, we used structure-guided mutagenesis followed by biochemical analysis. We chose residues S168, H170, S171 and H182 and introduced single alanine substitutions, as well as one double (S168A H170A) and one triple (S168A H170A H182A) variant. To ensure equal adenylation (that is activation) of all LIG variants, we preincubated the enzymes with ATP and performed gel-based analysis (Extended Data Fig. 3b). Next, we assessed RNA binding towards the ASL-4U substrate using EMSAs. The binding assays were conducted with non-hydrolyzable AMPcPP to prevent adenylation of the substrate,

which would otherwise 'anchor' it to the active site. Our analysis showed reduced binding of the S168A and H182A variants, as well as the double and triple mutants; H170A and S171A did not show a difference compared with the wild-type (WT) structure (Fig. 3f). We then determined ligation activity by monitoring time-dependent circularization of the ASL-4U substrate. The observed activities were similar to that of the WT ligase for all variants, except of the triple-mutant LIG, which reduced formation of the circular product (Extended Data Fig. 3c). The binding of the 5'- and 3'-exon ends is coupled in these experiments as they are both part of the same ASL-4U molecule. To uncouple binding of the 5'-exon end from that of its 3' counterpart, we performed separate strand-ligation assays of *HAC1*-mRNA-derived fragments, as established previously²². This revealed diminished ligation activity, in accordance with the EMSA results. Although the H170A and S171A variants showed almost the same ligation activity as the WT, formation of the ligated product was reduced for all other variants (Fig. 3g). Both the EMSA and separate strand-ligation data showed additive effects of combining two or three alanine substitutions in the proposed binding surface of the 5' exon. Taken together, our biochemical analyses support the importance of the identified residues in binding of the 5'-exon end and corroborate the phenotypes in yeast cells.

Molecular modeling of the LIG–ASL complex reveals CTD clamping

We propose a model in which the ends of the single-stranded RNA substrate engage with the LIG active site from opposite directions, while maintaining the base-pairing interactions of the stem (Fig. 3e). However, the important question of whether such coordination would be sterically possible in the context of a physiological pre-tRNA substrate was not answered by the crystal structure. To address this, we performed molecular modeling using specific AMBER⁴³ force fields^{44–47}. First, we generated a starting model by retaining the nucleotides from the proposed 5'- and 3'-exon equivalents with extensive protein contacts, as well as the base-paired stem, that is nucleotides 26–31 and 32–37 from two copies of the 5' exon, and nucleotides 38–44 from one copy of the 3' exon (Fig. 3c and Methods). After linking the backbone of nucleotides 31 and 32, we energy-minimized the conformation of the protein-unbound nucleotides 32–34 to yield a bound RNA stem loop with the same overhangs and stem as those in a cleaved pre-tRNA-ASL substrate. The ends of the starting model matched an activated (that is, adenylylated) physiological substrate (2'-P, 3'-OH and 5'-AppRNA). The resulting model showed that Trl1-LIG can accommodate a cleaved tRNA-ASL in its binding cleft as a hairpin structure (Fig. 4a). All base-pairing interactions in the hairpin remained intact, except for the first A-U base pair toward the loop, which was also unpaired in the starting model (Fig. 4b).

The molecular dynamics (MD) simulations revealed a conformational change that resulted in 'clamping' of the CTD onto the bound tRNA-ASL (Fig. 4c). We carried out Normal Mode Analysis (NMA) using the eInémo webserver⁴⁸ for the Trl1-LIG protein without RNA to identify potential conformational changes. NMA indicated that the clamping movement was the primary element of LIG dynamics (Fig. 4c and Supplementary Video 1). These results suggest that the NTD-to-CTD distance can adapt to accommodate different substrates. Compared with our crystal structure, in which CTD-mediated contacts with the bound RNA were largely absent, the modeling data revealed numerous residues in the CTD as a potential RNA-binding interface (Extended Data Fig. 6a). Many of these residues are conserved and have previously been shown to be essential in vivo⁴². Using principal component analysis, which is used to describe modes of protein motion, we found that the clamping motion of the CTD is also present in the Trl1-LIG–ASL complex (Extended Data Fig. 6b).

Last, we generated a tentative model of Trl1-LIG with tRNA to test the plausibility of its binding to a physiological substrate. We superimposed the anticodon stem of the yeast tRNA^{Phe} crystal structure

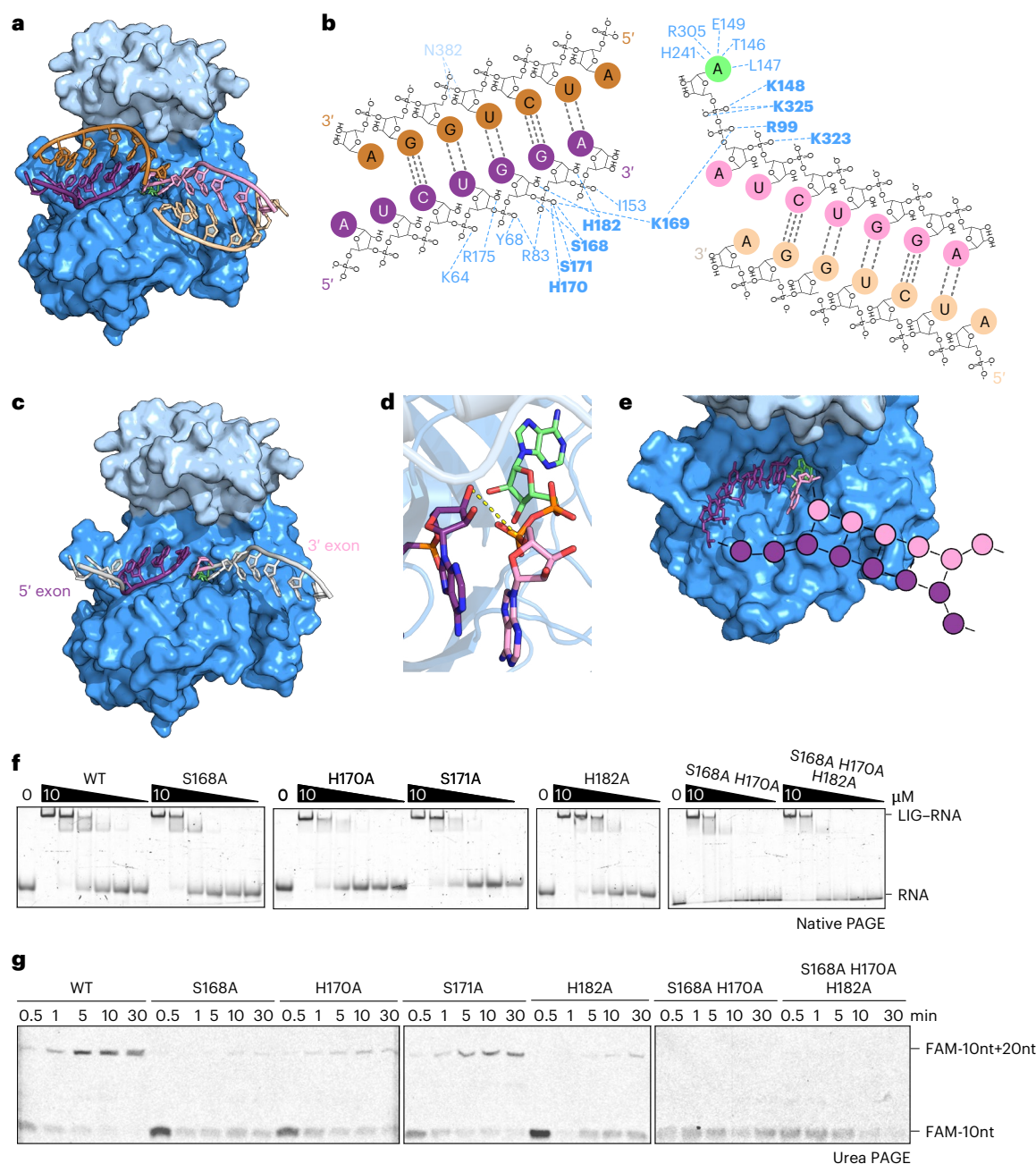


Fig. 3 | Analysis of the Trl1-LIG-RNA-binding interface. **a**, Surface representation of one *Ctr1*-LIG molecule with both interacting RNA duplexes in the crystal. The NTD is colored in blue and the CTD in light blue. The RNA strands (bronze and purple; sand and pink) are depicted as cartoons, with the AMP cofactor as sticks in lime. **b**, Interaction map of *Ctr1*-LIG residues with the RNA duplexes from the NTD and the CTD using the same color coding as in **a**. Essential residues are shown in bold. **c**, Representation of the exon-exon arrangement extracted from the *Ctr1*-LIG-RNA structure. The RNA exon strands are annotated as 5' exon (purple) and 3' exon (pink). RNA nucleotides without amino acid contacts are shown in white. **d**, Zoomed-in view of the *Ctr1*-LIG active site with the 3' end of the 5' exon (purple; red, oxygen; blue, nitrogen) and the activated 5' end of the 3' exon (pink; AMP in lime; red, oxygen; blue, nitrogen) depicted as sticks. The distance of 4.7 Å between the 3'-OH and the 5'-P is indicated by the yellow dashed line. **e**, Scheme of pre-tRNA ASL coordination. Nucleotide positioning

from the crystal structure is shown as sticks in the active site of *Ctr1*-LIG (blue). The extending nucleotides are depicted as circles indicating the ASL of a cleaved pre-tRNA (5' exon in purple and 3' exon in pink). **f**, EMSA of WT *Ctr1*-LIG and the S168A, H170A, S171A, H182A, S168 H170A and S168 H170A H182A mutants (0–10 μM). Increasing amounts of protein were incubated with the ASL-4U RNA (1 μM) in the presence of AMPcPP and analyzed by native PAGE using SYBRGold RNA staining. Formation of the LIG-RNA complex was assessed by band shift. This assay was performed in triplicate (*n* = 3). **g**, Fluorescence-based separate strand ligation assay with WT *Ctr1*-LIG and the S168A, H170A, S171A, H182A, S168 H170A and S168 H170A H182A mutants (100 nM). Time-course urea PAGE of the ligation of the 5' fluorescein (FAM)-labeled 10-nt HAC1 5' exon oligonucleotide (lower band) with a 20-nt HAC1 3' exon oligonucleotide was monitored using fluorescence detection. This assay was performed in duplicate (*n* = 2).

(PDB ID [1EHZ](#)) onto the protein-bound fragment of the ASL, and added the 'missing' part of the tRNA. This model demonstrated that binding of an entire tRNA is sterically compatible with the Trl1-LIG-ASL binding mode revealed by the crystal structure (Fig. **4d**, pink tRNA molecule).

We performed coarse-grained Monte Carlo dynamics simulations of the protein with the Trl1-LIG-tRNA complex, keeping the Trl1-LIG-NTD and nucleotides 35–38 of the ASL fixed, while allowing the Trl1-LIG CTD and the tRNA to move (Methods). The simulation suggested that the

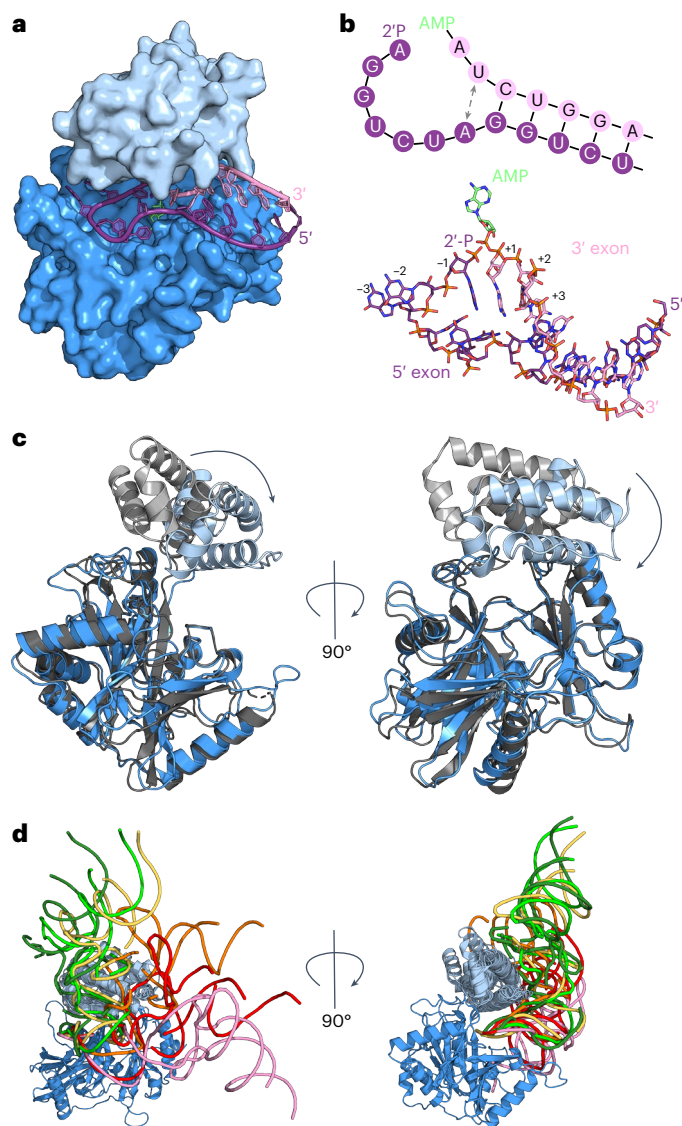


Fig. 4 | Molecular modeling of *CtrI*-LIG in complex with tRNA-derived substrates. **a**, Surface representation of the *CtrI*-LIG (blue) model with bound ASL exon halves (pink and purple). The model was derived from the MD simulation trajectory based on frames demonstrating the most stable RNA-protein interaction energies. **b**, Modeled structure of the ASL RNA in the active site with the 5'-end activated AppRNA 3' exon half (pink) and the 5' exon harboring a 2'P/3'OH end (purple). The structure is illustrated as a schematic (top) and as sticks (bottom, colored by element). The dashed line (gray) indicates the opened A-U base pair toward the loop. **c**, Cartoon representation of superposition with *CtrI*-LIG-RNA crystal structure (gray) and *CtrI*-LIG-ASL model (blue) without RNA depiction. The structures were superimposed along the core β -sheets of the adenylyltransferase domain. Superposition reveals CTD clamping, indicated by arrows towards the NTD active site. **d**, Cartoon representation of SimRNA models of *CtrI*-LIG with full-length tRNA. The starting model tRNA (pink) and medoids of the five largest clusters of best-energy conformations of the tRNA (dark green, light green, yellow, orange and red) are shown as ribbons.

tRNA molecule could adopt a range of possible conformations relative to LIG and confirmed clamping of the CTD (Fig. 4d).

Taken together, the results of molecular modeling confirmed the overall orientation and topology of the exon ends in the crystal structure, but suggested substantial conformational differences at the level of individual nucleotides. It further provided a model for the Trl1 engagement with physiological substrates, including the role of the unique CTD in RNA binding.

Conserved substrate-binding mode between adenylyltransferases

Up to now, the structure of T4 Rnl2 with a nicked nucleic acid substrate represented the only high-resolution snapshot of an RNA ligase with substrate³⁰. T4 Rnl2 repairs single-strand breaks in the context of double-stranded RNA. In this structure, the bound substrate is a DNA duplex with a terminal 2'-OH and an adenylylated 5' DNA end at the nick. It revealed two distinct states: (1) immediately after 5'-end adenylation (A form, step 2 product) and (2) before nucleophilic attack by the 3'-OH (B form, step 3 substrate). Despite acting on RNA substrates with different topologies and relatively low similarity (22% sequence identity, 34% sequence similarity) (Extended Data Fig. 7a), both enzymes belong to the same superfamily (CATH database 3.30.470.30: DNA ligase/mRNA capping enzyme superfamily). We used structural superposition of our LIG-RNA complex with the T4 Rnl2-substrate complex to compare substrate-binding modes. The three DNA strands in the Rnl2 structure, that is both nicked strands and the complementary template strand, have equivalents in our LIG-RNA structure (Fig. 5a). Notably, the nick 5' end is adenylylated and superimposes onto the AMP-3' exon strand in our structure. Both ligases utilize conserved lysine and arginine residues to coordinate the adenylylated 5' end (Fig. 5b). When comparing our structure with both states of Rnl2 with bound DNA, the strand harboring the 3' nick and the complementary template strand are superimposed with one bound RNA duplex in our structure (Fig. 5c). The adenine base and ribose of the cofactor nucleotide are superimposed as well. The position of the α P and the orientation of the phosphoanhydride bond are more similar to the post-step-2 state (complex A), in line with our interpretation. Superposition with human³² and *E. coli*³¹ DNA ligase structures with adenylylated DNA revealed a similar overall architecture of ligase-substrate complexes (Extended Data Fig. 7b–d). Taken together, comparisons of both RNA ligase-substrate complexes confirmed an RNA-duplex-binding interface and the specific coordination of both exon ends toward the catalytic center.

Role of the CTD during Trl1-LIG-mediated ligation

The CTD of Trl1-LIG adopts a unique fold^{21,22} and has been shown to be essential for ligase activity in vivo⁴⁹. Although our substrate complex structure identified limited contacts of residues from the CTD to the bound RNA, molecular simulations revealed additional RNA-binding residues in the CTD through clamping of the subdomain onto the RNA (Fig. 4c and Extended Data Fig. 6a). To further dissect the role of the CTD, we generated a truncated version of *CtrI*-LIG (Δ CTD; amino acids 1–328) and determined its effect on each catalytic step of the Trl1-LIG-catalyzed ligation reaction (Fig. 1a). We tested the correct folding of the truncated variant in a thermostability assay (Extended Data Fig. 8c) and analyzed the contribution of the CTD during the required adenylylation of Trl1-LIG (that is, the activation step). To this end, we monitored the formation of the covalent LIG-K148-AMP intermediate using radioactive [α -³²P]ATP with WT LIG, as well as the K148N and Δ CTD variants (Fig. 6a). WT LIG showed concentration-dependent adenylylation, which was completely absent for the K148N variant. The degree of adenylylation for the Δ CTD variant was indistinguishable from that of the WT protein, which we confirmed in a time-course assay (Extended Data Fig. 8a,b). Our data revealed that the CTD is dispensable during ligase activation.

We determined the substrate-binding capacity of the different LIG variants using EMSAs with the ASL-4U substrate (Extended Data Fig. 1b). Deletion of the CTD strongly reduced RNA binding compared with levels for the WT LIG and the K148N variant (Fig. 6b). Notably, the ASL-4U substrate harbors an adenine base at the 5' end. To estimate the influence of the base at the RNA end, the EMSA was repeated with the same RNA harboring a cytidine at the 5' end (ASL-4U-5C). The WT LIG showed higher RNA binding than did the K148N mutant (Extended Data Fig. 8d). For the Δ CTD variant, we observed detectable levels of binding

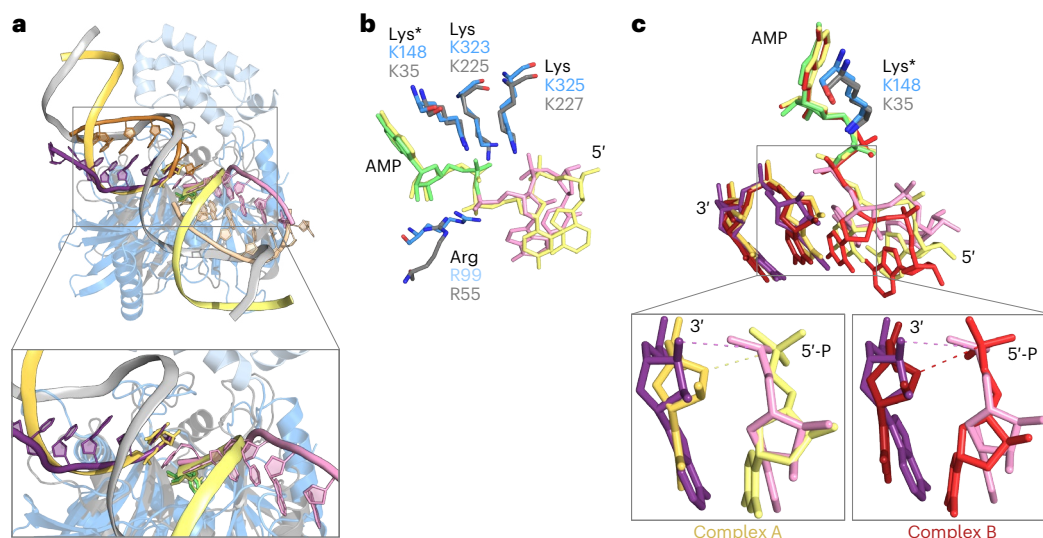


Fig. 5 | Conserved RNA binding modes between *CtrI1*-LIG-RNA and T4 Rnl2-AppDNA. **a**, Superposition of the *CtrI1*-LIG-RNA structure and the structure of T4 Rnl2 (gray) in complex with an AppDNA substrate (yellow, template strand in gray; PDB ID 2HVR, A form). The structures were superimposed along the core β -sheets of the adenyltransferase domain (r.m.s.d. = 1.043 Å). The DNA strands are depicted as cartoon strands. The rectangles show the enlarged view. Note the omission of one RNA strand (compare Fig. 3a, bronze strand). **b**, Zoomed in view of the activated 5' end of the nucleic acids after superposition of the structures from A depicted as sticks. The active site lysines (Lys*, K148 of *CtrI1*-LIG and K35 of T4 Rnl2) as well as the conserved coordinating lysines (K323 and K325 of

CtrI1-LIG, K225 and K227 of T4 Rnl2) and arginines (R99 of *CtrI1*-LIG and R55 of T4 Rnl2) are shown as sticks. **c**, Zoomed in view of the activated 5' end of the nucleic acids and the 3' end of the corresponding 5' exon after superposition of the structures from A as well as an alternative conformation of the T4 Rnl2 complex with an AppDNA substrate (red; PDB ID 2HVR, B form). The active site lysines are depicted as sticks (K148 of *CtrI1*-LIG and K35 of T4 Rnl2) together with the first 2 nt of both exon strands. The individual superpositions are depicted on the bottom (rectangles). Dashed lines indicate the distance of the 3' end to the 5'-P (pink, 4.7 Å; yellow, 4.4 Å; red, 3.6 Å).

only at a 100-fold-higher enzyme concentration than for the WT LIG. The EMSA results reveal that the CTD contributes to RNA binding.

During ligation, the LIG-bound AMP forms a covalent anhydride with the 5'-P end of the RNA, resulting in an AppRNA intermediate. To monitor the formation of this adenylated RNA intermediate, we performed ligation reactions with the ASL-4U RNA in the presence of [α^{32} P]ATP. The autoradiogram revealed signals for radioactively labeled RNA in the presence of WT LIG (Fig. 6c, lanes 2–5). No activated RNA intermediate was detected for the K148N (Fig. 6c, lanes 6–9) and Δ CTD (Fig. 6c, lanes 10–13) variants. With increasing reaction time, the signal for the AppRNA intermediate using WT LIG subsided owing to completion of the ligation reaction and concomitant release of AMP (Extended Data Fig. 8e,f). When using the ASL-4U substrate with the non-ligatable 3' end (2'-OH and 3'-P (2'-OH/3'-P) and 5'-P), we observed a continuous increase in adenylation over the same time-course (Extended Data Fig. 8c,d).

Finally, we tested the final ligation step in circularization assays. In the presence of WT LIG, the linear substrate was completely circularized within 30 min (Fig. 6d, lanes 2–5). Addition of Δ CTD or the ligase-dead K148N variant showed no activity under the same conditions (Fig. 6d, lanes 6–13). These biochemical assays revealed that the CTD is dispensable during ATP-dependent activation of TrlI-LIG. However, presence of the CTD promoted RNA binding and was required for transfer of the AMP moiety from the active-site lysine onto the 5' RNA end as well as the final ligation step. Taken together, in vitro analyses rationalize the importance of the CTD in vivo.

The CTD provides 2'-P specificity through conserved arginines

The requirement for a terminal 2' phosphate is a specific characteristic of TrlI-type RNA ligases^{5,50}. The 2' phosphate is not present in our crystal structure of LIG-RNA, but we can infer an approximate location of the 2'-P from the terminal 2'-OH end and its position near the active site (Fig. 7a). Notably, a sulfate ion occupies this position

in previous *CtrI1*-LIG structures^{21,22}. Superposition of the LIG-RNA structure and the LIG-AMPcPP structure (PDB ID 6N67) depicts the potential 2'-P pocket (Fig. 6b). The sulfate ion is coordinated by two conserved arginines in the CTD, R334 and R337 (Fig. 7b orange sticks, see also conservation scale in Extended Data Fig. 4), which are located near the RNA 3' end in our LIG-RNA structure (Fig. 7a). To assess the role of R334 and R337 in 2'-P specificity, we purified *CtrI1*-LIG variants R334A and R337A, as well as the double mutant R334A R337A, and determined their activity toward 2' positions and 3' RNA ends. We performed ligation assays using the ASL-4U substrate harboring a 5'-P in combination with a 2'-P and 3'-OH (the same as endogenous substrates), a 2'-OH/3'-OH or a 2'-OH/3'-P end. The assays revealed slower ligation kinetics for all three Arg-to-Ala variants towards the endogenous-like RNA in comparison to WT LIG. Reduced ligation kinetics of the variants coincided with an accumulation of the AppRNA intermediate (Fig. 7c, top). The observed effects for the single mutant variants were additive in the double mutant. Analysis of the AppRNA formation and the circularization kinetics confirmed the reduced rate constants for AppRNA formation of the arginine variants (Fig. 7d). The circularization assay with the 2'-OH/3'-OH and 5'-P ASL-4U substrate (containing the non-physiological 2'-OH end) resulted in diminished circularization activities with a slightly increased activity of the R334A variant compared to WT, R337A and R334A R337A (Fig. 7c, middle). The correct identity of the substrate ends in these assays was confirmed using T4 Rnl1, which did not ligate the 2'-P- and 3'-OH-containing ASL-4U, only its canonical 2'-OH- and 3'-OH-containing substrate. In a control experiment using the ASL-4U with 2'-OH/3'-P and 5'-P ends, none of the ligases exhibited circularization (Fig. 7c, bottom). We observed formation of the AppRNA intermediate for the 3' exon fragment when combined with non-productive 5' exon fragments albeit with reduced kinetics (Fig. 7c, middle and bottom, marked with *). Taken together, these results indicate a major preference, but not a strict requirement, for 2'-P-containing substrates by *CtrI1*-LIG, which is strongly dependent on the CTD residue R334.

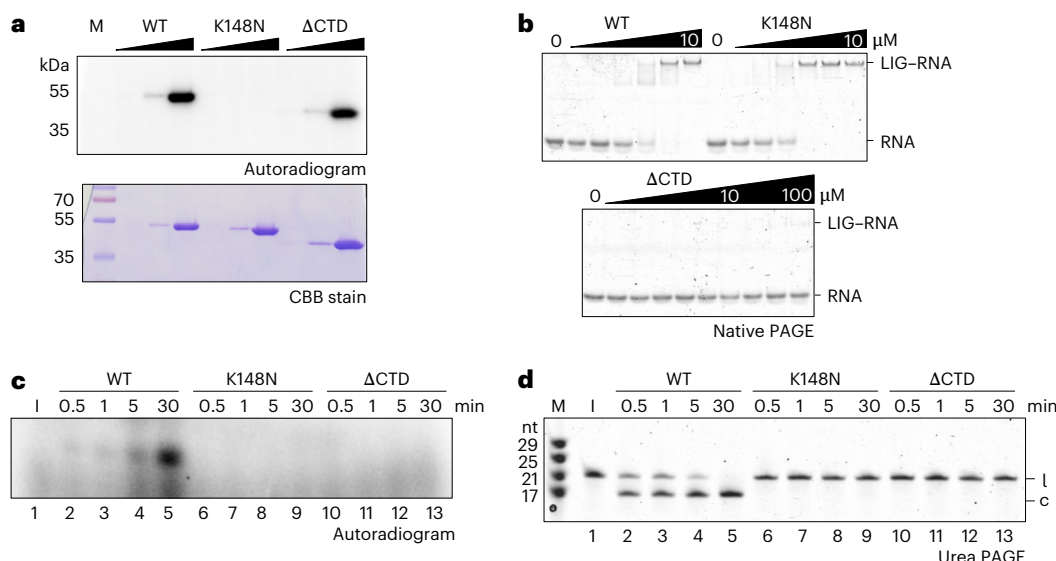


Fig. 6 | Role of the Trl1-LIG C-terminal domain during RNA ligation.

a, Adenylyltransferase assay of *CtTrl1*-LIG WT, K148N and Δ CTD variants. Top: the autoradiogram after incubation of increasing protein concentrations (0.1–10 μ M) with [α - 32 P]ATP. Bottom: the SDS–PAGE was stained with Coomassie brilliant blue (CBB) as protein loading control. This assay was performed in duplicate ($n = 2$). **b**, EMSA with WT *CtTrl1*-LIG and the K148N (both 0–10 μ M) and Δ CTD (0–100 μ M) variants. Increasing amounts of protein were incubated with the ASL-4U RNA (1 μ M) and analyzed by native PAGE using SYBRGold RNA staining. Formation of the LIG–RNA complex was assessed by band shift. This

assay was performed in triplicate ($n = 3$). **c**, RNA adenylation by WT *CtTrl1*-LIG and the K148N and Δ CTD variants (1 μ M). The autoradiogram shows formation of the adenylylated ASL-4U intermediate via [α - 32 P]AMP incorporation over time. The depicted gel represents the final result after validation of the assay in triplicate (Extended Data Fig. 8e, $n = 3$). **d**, Circularization assay of WT *CtTrl1*-LIG and the K148N and Δ CTD variants (1 μ M). Ligation of the linear ASL-4U substrate (l) to the circular form (c) was monitored by urea PAGE. This assay was performed in triplicate ($n = 3$).

Discussion

The fungal tRNA ligase Trl1 is an essential enzyme for both tRNA splicing and non-conventional *HAC1* mRNA splicing during the UPR^{5,12}. Despite recent progress, the principles of RNA substrate recognition by Trl1 remain unknown. Overall, our understanding of RNA ligases that join single-stranded RNA substrates is limited by the scarce availability of ligase–RNA structures. Here, we present the crystal structure of *CtTrl1*-LIG with an RNA substrate. Our data provide insights into coordination of the RNA exons by Trl1-LIG, including a conserved RNA-binding surface. We further identified molecular determinants for the specific biochemical mechanism.

By comparing our LIG–RNA structure with previously reported structures of RNA and DNA ligases with an activated nucleic acid substrate, we identified conserved binding for ATP-dependent activation of the 3' fragment (or exon). We surmise that orientation of the activated strand is mostly invariable between adenylyltransferases. Relative to the activated strand, the 5' fragment (exon) is coordinated in an apical orientation, allowing in-line attack of the 3'-OH nucleophile toward the 5'-end phosphorus atom, a hallmark of phosphoryl transfer reactions⁵¹. This substrate-binding mode engages the entire RNA-binding cleft of Trl1-LIG. In vivo lethality and reduced in vitro activity of RNA-binding-site mutants corroborate our findings. We propose a principle for RNA-substrate recognition and binding that is shared between nick-sealing adenylyltransferases (both RNA and DNA ligases) as well as RNA ligases joining single-stranded breaks.

The 2'-P requirement is a defining characteristic of Trl1-type ligases^{5,50}. Recent structural studies have suggested that two arginine residues in the CTD of fungal Trl1-LIG that coordinate a sulfate ion in place of a phosphate are determinants for the 2'-P specificity^{21,22}. The LIG–RNA structure proposes a position for the 2'-P on the bound 5' exon equivalent in the vicinity of the CTD arginines. We conclude that the Arg-to-Ala variants show an accumulation of the AppRNA intermediate due to perturbed positioning of the 3'-OH through the 2'-P interaction. Although the impact of p.R334A was stronger than

that of p.R337A in our in vitro experiments, yeast mutagenesis of the equivalent arginines revealed that only R337 is essential in vivo⁴². Conceivably, both arginines act together in 2'-P binding, thereby orienting the 3'-OH end in a position favorable for nucleophilic attack of the 5' end. However, the precise role of each arginine remains to be elucidated. Of note, both conserved arginines in fungi are absent in plant Trl1. In contrast to *Arabidopsis thaliana* Trl1, we observed a less strict 2'-P requirement of fungal Trl1-LIG for AppRNA formation⁵⁰. Given the lack of plant Trl1-LIG structures, we speculate that their CTD fold differs from that of their fungal counterparts. The presence of varying C-terminal subdomains is a common feature of adenylyltransferases. These C-terminal segments promote distinct specificity for their cellular substrates^{23–25}. We show that the CTD of Trl1-LIG is not required for ligase activation, but plays a role in RNA binding. Computational analyses suggest that the binding cleft between NTD and CTD 'clamps' onto the RNA ends, utilizing conserved RNA-binding residues in the CTD. We conclude that the importance of the CTD lies in stabilizing RNA binding and positioning of the 5' exon's 3'-OH end for ligation through 2'-P coordination. Additionally, CTD-tRNA contacts outside the ASL could promote specificity for pre-tRNA substrates, particularly in a cellular environment.

The RNA-binding residues contact the phosphate backbone rather than RNA bases, explaining why Trl1 can act on a variety of RNA substrates independently of their sequence, as long as they possess ends with a 2'-P/3'-OH and a 5'-P. This ability is reflected in the diversity of cellular substrates and virtually unrestricted activity towards in vitro substrates.

Circularization assays with different RNA substrates suggest that Trl1-LIG favors exon ends held in close proximity by double-stranded segments. Canonical substrates, SEN-cleaved pre-tRNAs and Ire1-cleaved *HAC1* mRNA exhibit exon–exon base-pairing^{34–36,38}. This RNA-intrinsic feature ensures joining of the correct exon ends, despite the presence of other RNA fragments in the cell and favors ligation over degradation by exonucleases^{22,52}. Our structural insights into

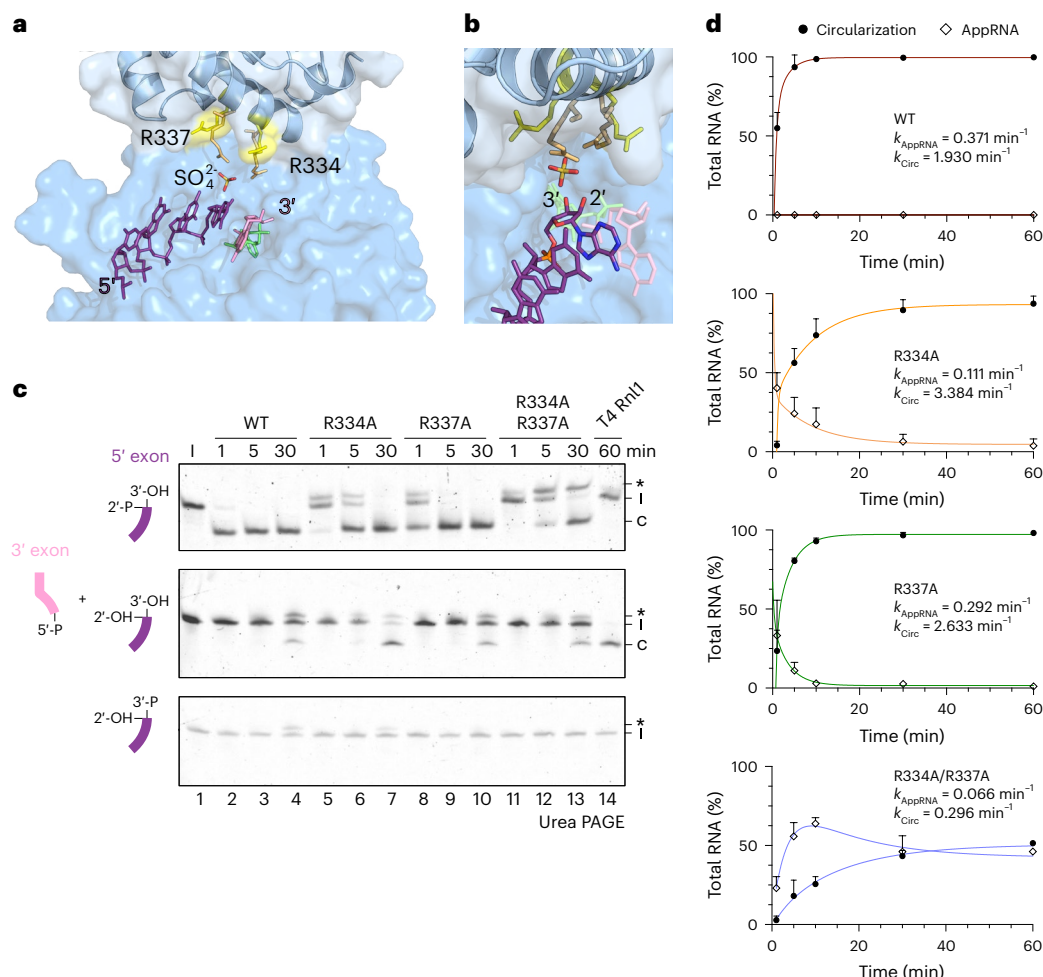


Fig. 7 | Conserved arginine residues within the CTD mediate 2'-P specificity.

a, The position of a bound sulfate ion (orange) from the previous crystal structure of *CtTrl1*-LIG in complex with AMPcPP (PDB 6N67) is shown in the context of the *CtTrl1*-LIG-RNA structure. Depiction of the sulfate ion based on the structural alignment of both crystal structures. The CTD residues R334 and R337 (yellow from 8RBJ or orange from 6N67) were previously found to coordinate the sulfate ion. The RNA strands (purple and pink) with the bound AMP (lime) are depicted as sticks. **b**, A zoomed-in view of the 3' RNA end coordination. The sulfate ion is located in a potential 2'-P pocket in close proximity to the 2'-OH end of the 5' exon end (purple, colored by element). **c**, Circularization assay of WT *CtTrl1*-LIG and the R334A, R337A and the R334A/R337A variants (1 μM) with the ASL-4U

substrate and the indicated end modification on the 2' and 3' ends of the 5' exon. T4 Rnl1 was used as ligation control (single timepoint). Ligation of the linear ASL-4U substrates (l) to the circular form (c) with the AppRNA intermediate (*) was monitored by urea PAGE. All assays were performed in triplicate ($n = 3$). **d**, Quantification of the AppRNA formation and circularization assay in **c**. The relative amounts of AppRNA and circularized RNA to the total RNA (%) are plotted as a function of reaction over time (min). Values represent the mean and s.d. of technical replicates ($n = 3$). The data in the graphs were fit as two-step reactions (Methods). The rate constants for the AppRNA formation (k_{AppRNA}) and the circularization reaction (k_{Circ}) are shown.

RNA coordination do not show binding of an entirely physiological substrate. Nevertheless, they provide insights into the coordination of both exon termini and identify important RNA-binding residues. Taken together, our LIG-RNA structure expands the collection of structural snapshots along the Trl1-LIG reaction mechanism.

Interdomain substrate transfer within the full-length, tripartite Trl1 remains unknown. How are both exon-ends passed to the LIG domain after modification by the KIN and CPD domain? Structural insights into the coordination of the RNA substrate by the KIN and CPD domains are currently missing, but would promote our understanding of the interplay between all three Trl1 active sites during catalysis.

Finally, Trl1 is often discussed as a potential target for antifungal therapy owing to fundamental differences compared with the human tRNA ligase complex. In this study, we provide insights on how to potentially perturb RNA binding or harness 2'-P specificity. In summary, our study uncovers conserved RNA binding modes between nick-sealing and single strand-end joining RNA ligases, indicating how different

RNA ligase families evolved from a common principle to perform their specific cellular tasks.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41594-025-01589-3>.

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Methods

Site-directed mutagenesis

The pET15b-*CtTrlI*-LIG plasmid²² served as the template for PCR-based site-directed mutagenesis. For the K148N and E328-stop (Δ CTD) variants, one oligonucleotide was used; two oligonucleotides and the Q5 Site-Directed Mutagenesis Kit (NEB) were used to create the S168A, H170A, S171A, H182A, R334A and R337A variants and the R334A R337A double variant. Oligonucleotides used for mutagenesis are listed in Supplementary Table 1. Positive constructs were transformed in *E. coli* BL21-CodonPlus (DE3)-RIPL (Agilent Technologies) and used for protein expression, as described below. Plasmids harboring the S168A H170A and S168A H170A H182A mutations were synthesized (BioCat).

Expression and purification of WT *CtTrlI*-LIG WT and variants

In this study, recombinant His₆-tagged *CtTrlI*-LIG (residues 1–414) and its variants were expressed in *E. coli* BL21-CodonPlus (DE3)-RIPL (Agilent Technologies). Cells were grown in Luria broth medium containing ampicillin at 37 °C until an optical density at 600 nm of 0.8 was reached. Expression was induced by 1 mM isopropyl- β -D-galactopyranoside (IPTG), and cells were grown for 3 h at 30 °C and collected by centrifugation. The cells were lysed in lysis buffer (25 mM HEPES/NaOH pH 7.5, 500 mM NaCl, 1 mM DTT, 25 mM imidazole, 1× complete protease inhibitor cocktail (Roche), and 10 mM Na₂P₂O₇ for de-adenylylated state purification) using a microfluidizer. Purification was performed by Ni²⁺ affinity chromatography using a HisTrap FF column (GE Healthcare Life Sciences) in elution buffer (25 mM HEPES NaOH pH 7.5, 500 mM NaCl, 1 mM DTT, 500 mM imidazole). After Tag removal in cleavage buffer (25 mM HEPES NaOH pH 8.0, 150 mM NaCl, 2.5 mM CaCl₂) by thrombin protease (Cytiva), the protease was removed from (un)cleaved target protein using a HisTrap FF column (GE Healthcare Life Sciences). Remaining nucleic acids were removed using anion exchange chromatography (AEX) on a HiTrap Q HP 5-ml column with a NaCl gradient in high salt buffer (25 mM HEPES NaOH pH 8.0, 1 M NaCl, 1 mM DTT). Purification continued through size-exclusion chromatography (SEC) on a HiLoad 16/60 Superdex200 pg column (GE Healthcare Life Sciences) in SEC buffer (10 mM HEPES/NaOH pH 7.5, 150 mM NaCl, 0.5 mM TCEP). Proteins were concentrated using Amicon Ultra-15 Centrifugal Filter Units with a 30-kDa molecular weight cut-off (Merck).

His₆-tagged *CtTrlI*-CPD (residues 576–846) and His₆-tagged *E. coli* RtcA were expressed and purified as described previously²².

Crystallization

For crystallization of His₆-*CtTrlI*-LIG in complex with an RNA oligonucleotide (5'-P-AUCUGGA-3'), protein (final concentration of 5.8 mg ml⁻¹) and RNA were mixed in a 1:1.7 (protein to RNA) ratio and used for sitting drop vapor diffusion at 291 K. Microcrystals appeared after 10 days in 0.2 M potassium formate and 20% PEG 3350. Microcrystals were flash-frozen in liquid nitrogen using 20% ethylene glycol as a cryoprotectant.

Data collection, structure determination and analyses

Data for the His₆-*CtTrlI*-LIG-WT–RNA duplex were collected at cryogenic temperature on the ESRF beamline ID23-2, using the MeshAndCollect workflow⁵³. This process involved integrating diffraction images from multiple small crystals within a single loop using XDS⁵⁴. The data were then clustered, combined and finally scaled using AIMLESS⁵⁵ as part of the CCP4i software package⁵⁶. The resolution cut-offs were selected on the basis of the half-data set correlation ($CC_{1/2}$), as implemented in AIMLESS⁵⁷. Phases were obtained by molecular replacement with PHASER-MR⁵⁸ implemented in the PHENIX package⁵⁹. A previously solved structure of *CtTrlI*-LIG (residues 1–414; PDB ID 6N67) served as the search model. Iterative model building and refinement was performed with Coot⁶⁰ and Phenix.refine⁶¹. The

quality of the resulting structural models was analyzed with MolProbity⁶². Crystallographic data are summarized in Table 1. Coordinates and structure factors are deposited at the Protein Data Bank PDB with accession code 8RBJ.

Structures were prepared with PyMOL 2.4.1 (The PyMOL Molecular Graphics System, Schrödinger). RNA–ligand interactions and binding surfaces were analyzed using PDB ePISA⁶³, PLIP⁶⁴ or LigPlot+ (ref. 65).

Molecular simulations

TrlI-LIG residues 56–60, which were unresolved in the structure of the TrlI–LIG complex with the activated RNA substrate (PDB ID 8RBJ), were modeled on the basis of the structure of the TrlI-LIG-Mn²⁺ soak structure (PDB ID 6NOV). The RNA duplex was used as a starting point to model the tRNA^{Phe} anticodon loop in the cleaved and activated state. The structure comprising two fragments ²⁶UCUGGAucugga³⁷, and AMP³⁸AUCUGGA⁴⁴ was generated, in which the uppercase nucleotides were taken from two strands of the duplex, and the lowercase nucleotides were taken from chain B of the symmetry-related molecule. A37 was converted to adenosine-2'-5'-diphosphate (A2P). The gap between nucleotides 31 and 32 was sealed by energy-minimizing the conformation of nucleotides 29–31 until convergence, using the conjugate gradient approach implemented in Swiss PdbViewer⁶⁶ with standard convergence criteria and force field parameters.

The simulations employed the Amber ff99SB force field with χ OL3 corrections for RNA^{44,46,47} and Amberff14SB⁴⁵ for protein. Parameters for AMP and A2P, treated as modified nucleotides within the RNA molecule, were generated using the Generalized Amber Force Field⁴⁵ and the Antechamber module^{67,68} available in the AMBER22 (ref. 43) molecular dynamics package. Explicit solvation was modeled using the TIP3P⁶⁹ water model, and the system was neutralized by adding an appropriate number of Na⁺ ions. To accommodate periodic boundary conditions, the entire system was positioned within a cubic simulation box, ensuring a minimum distance of 12 Å between the system's edge and the box boundaries. Initially, energy minimization was performed on the solvated systems to alleviate any potential steric clashes between atoms. Following minimization, a multi-stage equilibration procedure was implemented. First, the systems underwent heating from 0 to 300 K under constant volume conditions for 100 ps. During this stage, positional restraints of 10 kcal per mol Å² were applied to the backbone atoms of both the protein and RNA molecules. Subsequently, the temperature was further elevated to 300 K through three consecutive ~200-ps MD simulations under isothermal–isobaric conditions. The restraints were gradually reduced to 1 kcal per mol Å (ref. 46) during these NPT simulations. Finally, all restraints were removed, and 1-ns MD simulations were performed under both NPT (constant pressure) and NVT (constant volume) conditions. In the NPT simulations, a pressure of 1.0 atm was maintained using the Berendsen barostat⁷⁰ with a relaxation time of 2 ps. The SHAKE⁷¹ algorithm was employed to constrain all bond lengths involving hydrogen atoms. Short-range electrostatic and Lennard–Jones interactions were calculated using a 10-Å cutoff, and the Particle Mesh Ewald summation⁷² method was used to account for long-range electrostatic interactions. A constant temperature of 300 K was maintained using Langevin dynamics⁷³ with a collision frequency (γ) of 1 ps⁻¹. The production phase of the simulation was then carried out in the NPT ensemble for 500 ns. All simulations were performed using the GPU-accelerated version of the AMBER22 (refs. 43,74) simulation package. Postsimulation analyses, including r.m.s. deviation, r.m.s. fluctuation and interaction energy calculations, were conducted using the cpptraj⁷⁵ module and the LIE⁷⁶ utility available in AmberTools22 (ref. 43). Interaction energies were determined by summing the pairwise electrostatic and van der Waals interaction energies between selected residue pairs.

Principal component analysis was conducted on the simulation trajectory of the RNA–protein complex using cpptraj⁷⁵ in the AmberTools22 (ref. 43) software suite.

Analysis of conformational changes in CtrI1-LIG and CtrI1-LIG–RNA

To assess the flexibility and preferred movements of the CtrI1-LIG protein, we used the eINémo web server for the NMA of macromolecules. This tool computes low-frequency normal modes, providing insights into potential conformational changes and dynamic properties of proteins⁴⁸. We used default parameters, with the exception of the range of the moves, which we increased to 150% of the default value.

A model of CtrI1-LIG with the complete tRNA molecule was generated by combining the model of the CtrI1-LIG–ASL complex with the crystal structure of yeast tRNA^{Phe} (PDB ID 1EHZ)⁷⁷. In brief, nucleotides 26–29 and 41–44 in the anticodon step of the RNA from both structures were superimposed, and the fragment of nucleotides 28–42 from the CtrI1-LIG–ASL structure was retained and combined with the rest of the complete tRNA structure. The complete model was generated by ModeRNA⁷⁸. For the simulations, we used SimRNA, a coarse-grained computational method for RNA-folding simulations and three-dimensional (3D) structure prediction⁷⁹. The CtrI1-LIG catalytic domain and the key interacting RNA residues of the anticodon loop (35–38) were restrained to retain the conformation observed in the crystal structure. The CtrI1-LIG CTD was allowed to move within a distance of 15 Å from the starting conformation. Additional restraints were used to retain the internal 3D structure of the protein domains and of the tRNA molecule. Simulations were carried out with default parameters, and to identify preferred conformations, clustering analysis was done for the 5% of best-energy frames within the r.m.s. deviation threshold of 3 Å.

Radioactive adenylation assay

The adenylation reaction was performed with 0.1, 1 or 10 µM protein, 10 nM ATP and 18.5 MBq ml^{−1} [³²P]ATP in assay buffer (20 mM HEPES NaOH pH 7.5, 70 mM NaCl, 2 mM MgCl₂, 1 mM TCEP, 5% glycerol) at 30 °C. The reaction was stopped at different timepoints in 4× loading dye (1 M Tris, pH 6.8, 350 mM SDS, 50% glycerol, 3.7 mM bromophenol blue, 25% β-mercaptoethanol), separated on a 12% SDS–PAGE, fixed in fixation solution (20% ethanol, 7% acetic acid), incubated overnight on a radioactivity detecting film and scanned using a FLA 7000 (FUJIFILM) or a Sapphire FL (Azure Biosystems).

Electrophoretic mobility shift assay

EMSAs were used to detect protein–RNA interaction complexes. RNA (1 µM) was mixed with increasing concentrations of protein (0.04–10 µM or 25–100 µM) and 100-fold excess of ATP or AMPcPP. Samples were incubated for 10 min at 30 °C and stopped in native loading dye (0.5× TBE, 5 mM EDTA, 5% glycerol, trace amounts of xylene cyanol and bromophenol blue). Samples were loaded on a 10% non-denaturing PAGE and run in 0.5× TBE buffer for 45 min at 100 V and 4 °C. The gels were subsequently stained with SYBR Gold nucleic acid stain (Invitrogen) to visualize the RNA-containing complexes by ultraviolet trans-illumination.

Radioactive RNA adenylation detection

To detect adenylylated RNA intermediate circularization assay was performed in presence of radioactively labeled ATP. The yeast tRNA^{Phe} stem-loop derived RNA substrate ASL-4U (Supplementary Table 2) was diluted in RNA buffer (20 mM HEPES NaOH pH 7.5, 100 mM NaCl and 1 mM MgCl₂), thermally unfolded at 90 °C for 1 min and refolded at room temperature by cooling down to −40 °C. Before circularization, the 3′ end of the RNA was end-modified by incubating 2 µM of RNA with 0.5 µM RtcA, 0.5 µM CtrI1-CPD, 10 nM ATP and 18.5 MBq ml^{−1} [³²P]ATP in RNA assay buffer (20 mM HEPES NaOH pH 7.5, 70 mM NaCl, 2 mM MgCl₂, 1 mM TCEP, 5% glycerol) for 30 min at 30 °C. Addition of 1 µM CtrI1-LIG or one of the variants initialized the reaction, resulting in adenylation of the RNA for the circularization of the stem-loop RNA over time. The reaction was quenched in 10-fold excess of stop solution

(10 M urea, 0.1% SDS, 1 mM EDTA, trace amounts of xylene cyanol and bromophenol blue). RNA samples were unfolded at 80 °C for 2 min. The samples were separated on a 15% denaturing urea PAGE. The gel was incubated overnight on a radioactivity detecting film and scanned using a FLA 7000 (FUJIFILM).

In vitro circularization assay

The ASL-4U RNA substrate and variants as well as the HAC1-4U substrate (Supplementary Table 2) were used for circularization to detect ligation activity of CtrI1-LIG-WT and variants. The RNA was diluted in RNA buffer (20 mM HEPES NaOH pH 7.5, 100 mM NaCl and 1 mM MgCl₂), thermally unfolded at 90 °C for 1 min and refolded at room temperature by cooling down to −40 °C. Before circularization, the 3′ end of the RNA was end-modified by incubating 2 µM of RNA with 0.5 µM RtcA, 0.5 µM CtrI1-CPD and 0.01 mM ATP in RNA assay buffer (20 mM HEPES NaOH pH 7.5, 70 mM NaCl, 2 mM MgCl₂, 1 mM TCEP, 5% glycerol) for 30 min at 30 °C. To ensure correct folding after the modification reaction, the truncated RNA substrates were refolded again. Addition of 1 µM CtrI1-LIG or variants initialized the ligation reaction, resulting in circularization of the stem-loop RNA over time. The reaction was quenched in 10-fold excess of stop solution (10 M urea, 0.1% SDS, 1 mM EDTA, trace amounts of xylene cyanol and bromophenol blue). RNA samples were unfolded at 80 °C for 2 min and subsequently placed on ice to prevent refolding. The samples were separated on a 15% or 12.5% denaturing urea PAGE and visualized through SYBR Gold nucleic acid stain (Invitrogen).

The relative amounts of AppRNA formation or circularized RNA were quantified relative to the total RNA (%) using ImageJ (Fiji). Data were plotted and analyzed using Prism10 software (GraphPad). The data in the graphs were fitted as two-step reactions using a biexponential association model with three constant terms *A*, *B* and *C* as well rate constants *k* and *k*₂ using the equation $y = A + B^{(-k \times x)} + C^{(-k_2 \times x)}$. *k*_{AppRNA} and *k*_{circ} were calculated by Prism10 (GraphPad).

Fluorescence-based separate strand ligation assay

HAC1-derived RNA oligonucleotides with a 10-nt-long 5′ exon fragment with a terminal fluorescein (FAM) fluorophore and a 20-nt-long 3′ exon fragment (Supplementary Table 2) were used as described previously²². One hundred nanomolar CtrI1-LIG WT or a variant was incubated with 250 nM of 3′ exon oligonucleotide and 0.5 mM ATP in RNA assay buffer for 15 min at 30 °C. The ligation kinetics were initiated by the addition of 50 nM FAM-labeled 5′ exon fragment. Aliquots were taken at the indicated timepoints and mixed with a threefold excess of stop solution. The samples were separated on a 15% or 12.5% denaturing urea PAGE and visualized and imaged for FAM fluorescence using a Amersham Imager (GE Lifesciences).

Yeast strains and growth conditions

All yeast strains in this study were based on *S. cerevisiae* W303, which had its *TRP1* and *ADE2* auxotrophy repaired, and are listed in Supplementary Table 6. Yeast transformations utilized the standard lithium acetate–single-stranded carrier DNA–PEG method. Cells were recovered on YPD medium and grown on the appropriate selection medium. To assess the impacts of several Trl1-LIG substitutions on yeast cell survival, the *trl1Δ* strain, rescued by plasmid-based, constitutive expression of CtrI1 under the ADH promoter²². We performed a plasmid shuffle using different pRS313-ScTrl1-3×HA constructs. The shuffle vector was cloned using Gibson assembly (NEBuilder Kit) after PCR amplification (Supplementary Table 3) of three DNA fragments with complementary overhangs: ScTrl1-3×HA (from a synthetic gene block) as well as 1 kb upstream and downstream of the *TRL1* gene (from genomic DNA). The fragments were cloned into the pRS313 vector⁸⁰ using restriction site cloning with EcoRI and SmaI. The 3×HA tag was introduced after the C-terminal domain of Trl1 for coimmunoprecipitation experiments not shown in this study. Substitutions in the

pRS313-ScTrlI-3×HA were introduced by site-directed mutagenesis (Supplementary Table 4) and confirmed by sequencing. The *trlIΔ* strains rescued by the p416ADH-CtTrlI plasmid were transformed with the respective shuffle vectors (Supplementary Table 5). Counterselection on 5-fluoroorotic acid (FOA) containing SDC plates was used to test for growth of the yeast strains with the introduced mutations in the *TRL1* gene.

For growth tests under different selection conditions, fivefold serial dilutions were spotted onto either standard synthetic dextrose complete (SDC)-Ura, SDC-His or SDC FOA plates. Images are shown in Extended Data Figure 5.

Nano differential scanning fluorimetry

Melting temperatures (T_m) of CtTrlI-LIG or the variants (50 μ M + 500 μ M ATP) were measured using nano differential scanning fluorimetry (nanoDSF). Intrinsic tyrosine and tryptophan fluorescence at emission wavelengths of 330 nm and 350 nm were measured continuously, applying a temperature gradient of 20–90 °C in the Prometheus Panta system (NanoTemper Technologies). The T_m was calculated using PR.PantaAnalysis software (Version 1.8, NanoTemper Technologies).

Denaturing polyacrylamide gel electrophoresis

RNA ligation and activation reactions were carried out as described above, and the reaction products were loaded onto a pre-run (300 V, 100 min) 12.5% or 15% urea gel. The gels were run in 1× TBE buffer at 150 V (constant voltage) at room temperature for 70 min. The gels were subsequently stained with SYBR Gold nucleic acid stain (Invitrogen) to visualize the RNA by ultraviolet trans-illumination.

Native polyacrylamide gel electrophoresis

EMSA samples were prepared as described above and loaded on a 10% native gel. The gels were run in cold 0.5× TBE buffer at 100 V (constant voltage) at 4 °C for 45 min. The gels were subsequently stained with SYBR Gold nucleic acid stain (Invitrogen) to visualize the RNA by ultraviolet trans-illumination.

Sodium dodecyl sulfate polyacrylamide gel electrophoresis

Adenylation reactions were carried out as described above. The reaction products were loaded onto a 12% SDS gel using 4× loading dye (1 M TRIS, pH 6.8, 350 mM SDS, 50% glycerol, 3.7 mM bromophenol blue, 25% β -mercaptoethanol). The gels were run in 1× SDS running buffer (248 mM TRIS base, 1.92 M glycine, 1% SDS) at 250 V (constant voltage) at room temperature for 35 min or in 1× MOPS-SDS buffer (50 mM MOPS, 50 mM TRIS, 0.1% SDS, pH 7.7). The gels were subsequently stained with Coomassie brilliant blue to visualize the proteins and were destained with 10% acetic acid solution.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Structure factors and atomic coordinates of the reported X-ray crystallographic structure have been deposited in the Protein Data Bank (PDB) under accession code [8RBJ](#). We have also compared and superimposed our structural models with the following PDB entries: [6N67](#), [6NOV](#), [2HVR](#), [2OWO](#), [1X9N](#) and [1EHZ](#). The MD simulation trajectory⁸¹ as well as the top elastic modes of NMA⁸² and the cluster representative for SimRNA simulation⁸³ are accessible at Zenodo. The data supporting the findings of this study are available within the paper and its Supplementary Information files. Should any raw data files be needed in another format, they are available from the corresponding author upon reasonable request. Source data are provided with this paper.

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Author contributions

S.K. performed experiments and analyzed data. J.K. and J.P. analyzed crystallographic data. S.M. and J.M.B. performed computational simulations. J.P. conceived the study and planned the experiments. The paper was written by S.K. and J.P. with contributions from all authors.

Competing interests

The authors declare no competing interests.

Additional information

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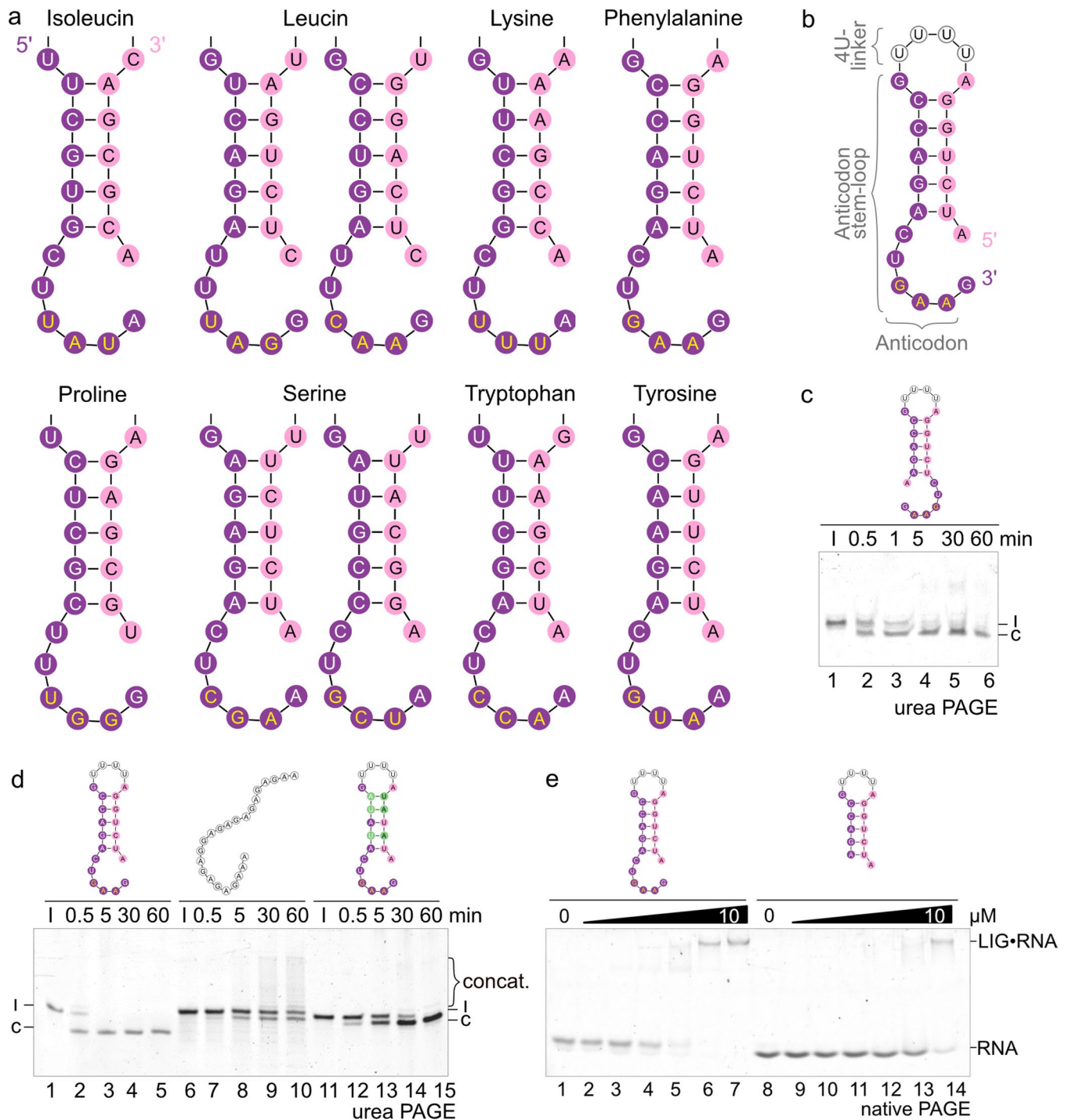
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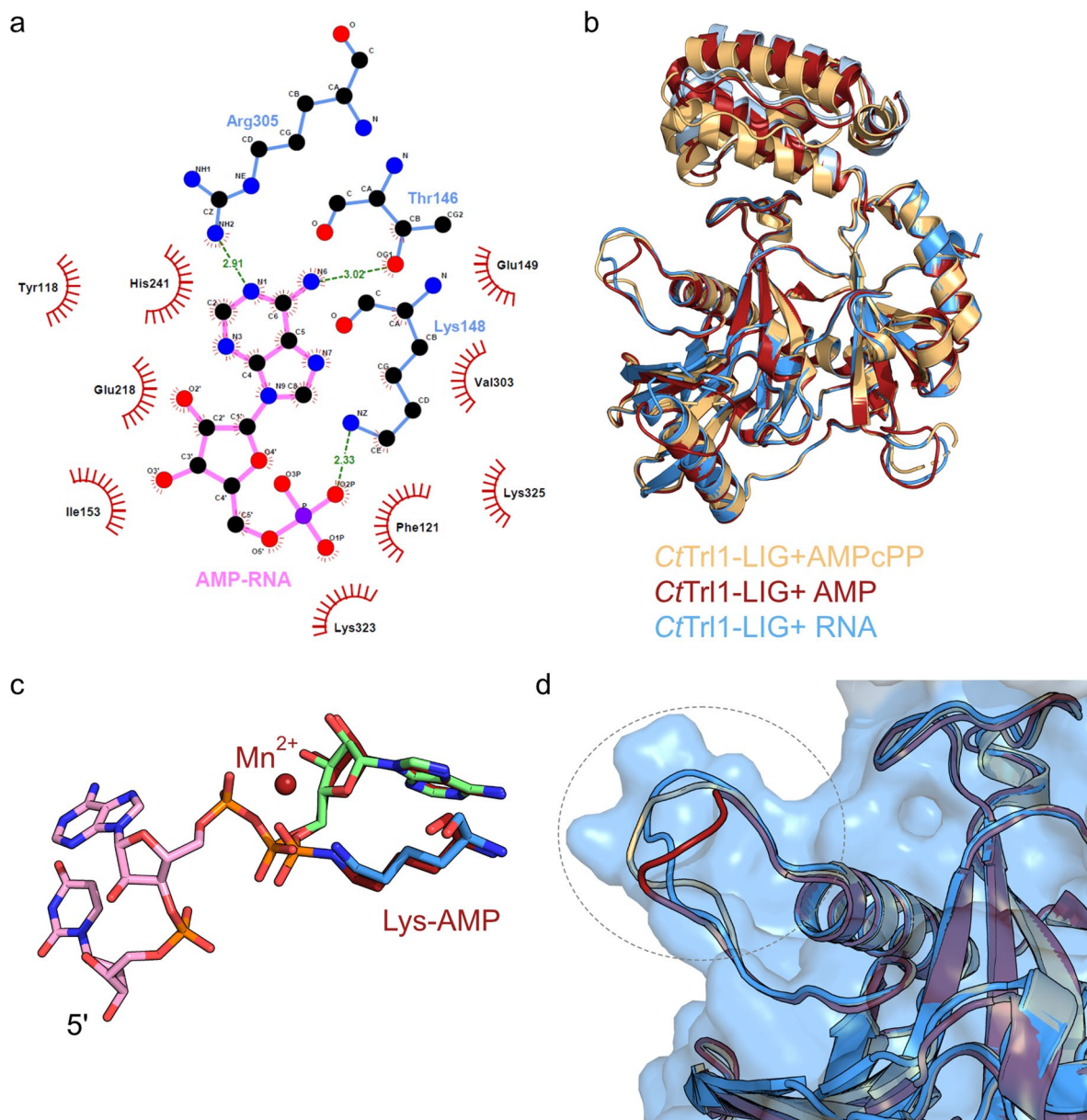
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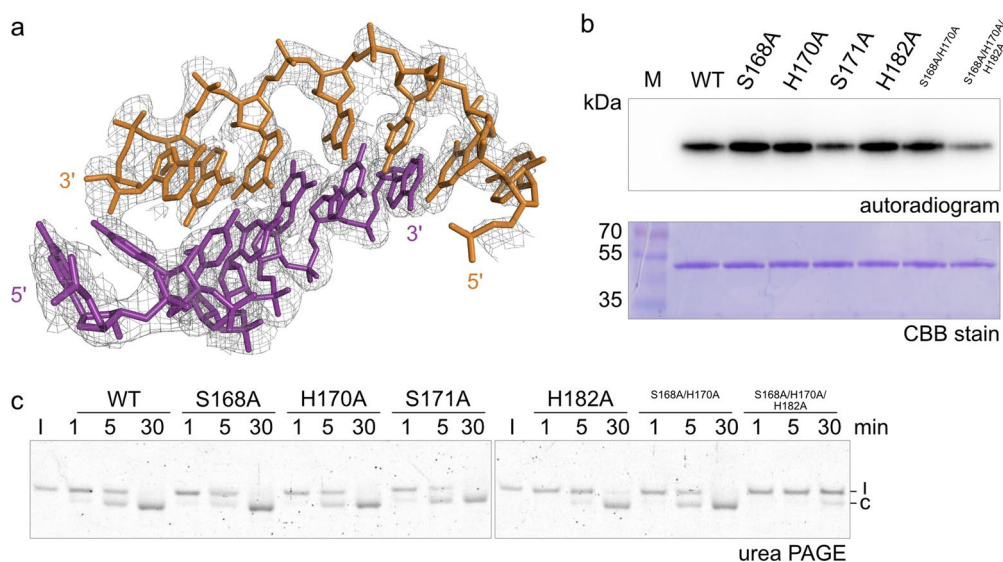
Extended Data Fig. 1 | Structures of TSEN cleaved pre-tRNAs, circularization assay substrate and RNA stem-loop structure determine *CtTrlI*-LIG activity. **a**, Yeast tRNA anticodon stem-loop structures. Overview of the anticodon stem-loop regions of all intron-containing tRNAs in yeast. The 5' exon is colored in purple and the 3' exon in pink. The sequences for the individual tRNA anticodon stem loops are depicted with the corresponding anticodon in yellow letters. **b**, Scheme of the ASL-4U RNA substrate for circularization, adenylation and binding assays. The RNA is derived from the TSEN-cleaved yeast tRNA^{Phe} anticodon stem-loop (ASL) with both strands linked by four uridines (4U). **c**, Circularization assay of WT *CtTrlI*-LIG (0.1 μM) with structured and unstructured

RNA substrates. Ligation of the linear ASL-4U substrates and the ASL-4U substrate with flipped ends (l) to the circular form (c) was monitored by urea PAGE. **d**, Circularization assay of WT *CtTrlI*-LIG (1 μM) with structured and unstructured RNA substrates. Ligation of the linear ASL-4U substrates (l) to the circular form (c) was monitored by urea PAGE. The bracket marks formed concatemers (concat.). **e**, EMSA with WT *CtTrlI*-LIG (0–10 μM) and ASL-4U or ASL-4U^{6nt} (1 μM). Increasing amounts of proteins were incubated with the RNA and analyzed by native PAGE. Formation of the LIG•RNA complex was assessed by band shift. All assays were performed in triplicates (n = 3).



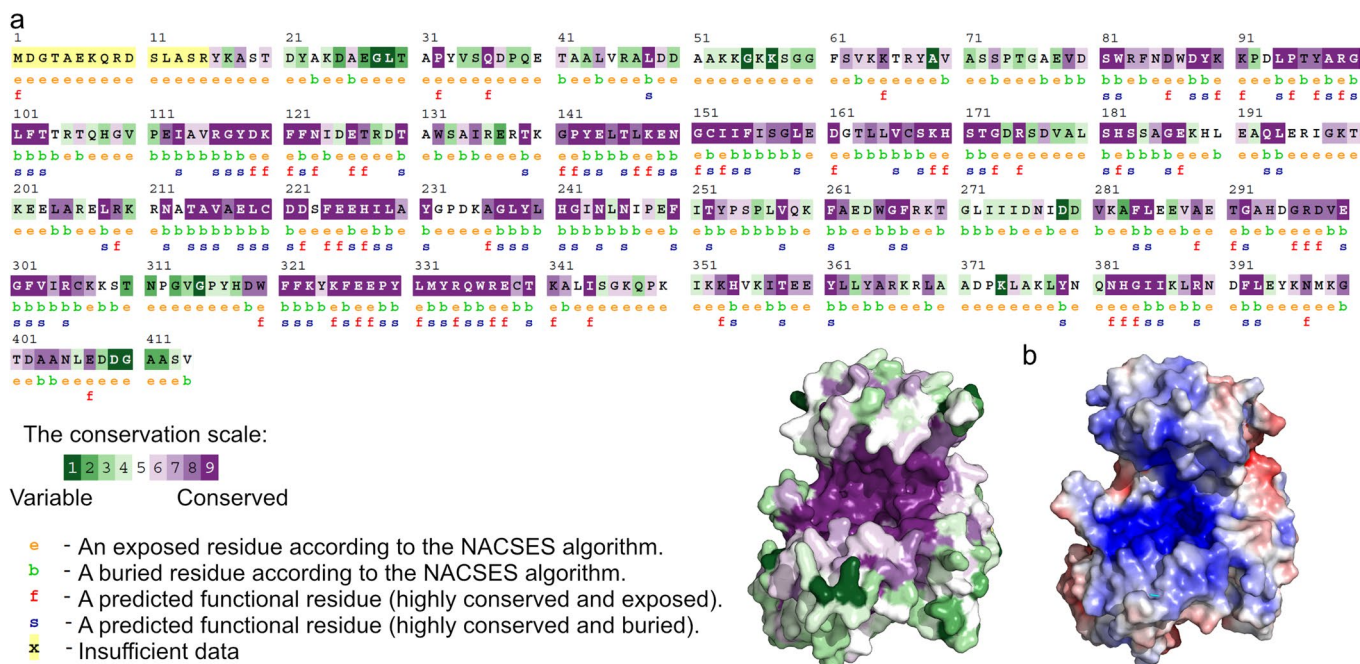
Extended Data Fig. 2 | 2D representation of CtrI1-LIG interacting residues with the AMP moiety of the activated RNA strand and structural alignment of CtrI1-LIG-RNA with other CtrI1-LIG structures. **a**, The bound AMP moiety (pink) is coordinated by hydrogen bonds (green dashed lines, number indicating the bond length) of CtrI1-LIG residues (blue). Non-ligand residues involved in hydrophobic contact(s) are labeled in black and shown as red fan. Atoms are colored and labeled by element. Created with LigPlot+¹⁵⁹. **b**, Superposition of CtrI1-LIG domain crystal structures in complex with AMPcPP (orange, PDB

ID: 6N67), AMP (red, PDB ID: 6NOV) and with RNA (blue, PDB ID: 8RBJ). The structures were superimposed along the β -sheets of the adenylyltransferase domain ($\beta 5$, $\beta 6$, $\beta 8$, $\beta 9$, $\beta 14$ and $\beta 15$). **c**, Zoom into the position of the active site lysine K148N relative to AMP moiety and activated RNA end of the CtrI1-LIG-AMP structure (red) and the CtrI1-LIG-RNA structure. **d**, Zoom into the thumb-like loop area (S168-S181) in the superposition of CtrI1-LIG structures. The surface of the CtrI1-LIG-RNA is presented transparently. The dashed circle marks the thumb-like protrusion. The color code is the same as in **a**.



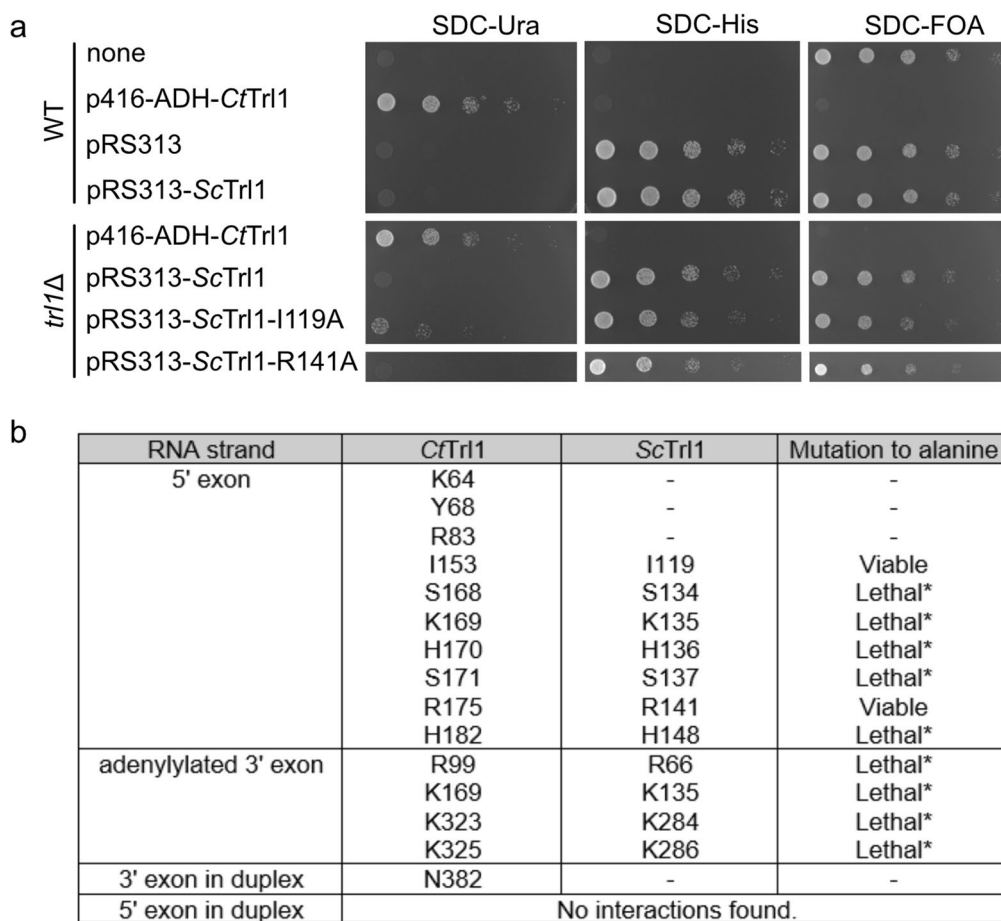
Extended Data Fig. 3 | Electron density map of the RNA duplex and in vitro activity assays of *CtrI1*-LIG mutants coordinating the 5' exon. **a**, Zoom into the RNA duplex forming strands (bronze and purple) depicted as sticks. The 2Fo-Fc electron density OMIT map at 1 σ is shown as a grey mesh. **b**, Adenylyltransferase assay of *CtrI1*-LIG WT and variants. The upper panel shows the autoradiogram after incubation of increasing protein concentrations (10 μ M) with α P³²-labeled

ATP. The SDS-PAGE was Coomassie Brilliant Blue (CBB) stained (lower panel) as protein loading control. This assay was performed once ($n = 1$). **c**, Circularization assay of *CtrI1*-LIG WT and variants (1 μ M). Ligation of the linear ASL-4U substrate (l) to the circular form (c) was monitored by urea PAGE. This assay was performed in duplicates ($n = 2$).



Extended Data Fig. 4 | Conservation analysis of the TrlI-LIG domain and electrostatic surface representation. a, The degree of amino acid conservation was determined using the ConSurf server⁴⁰. The *CtTrlI* sequence was used as input to calculate conservation of the residues in the LIG domain among TrlI-type ligases (with all parameters set to default). The conservation scale indicates the

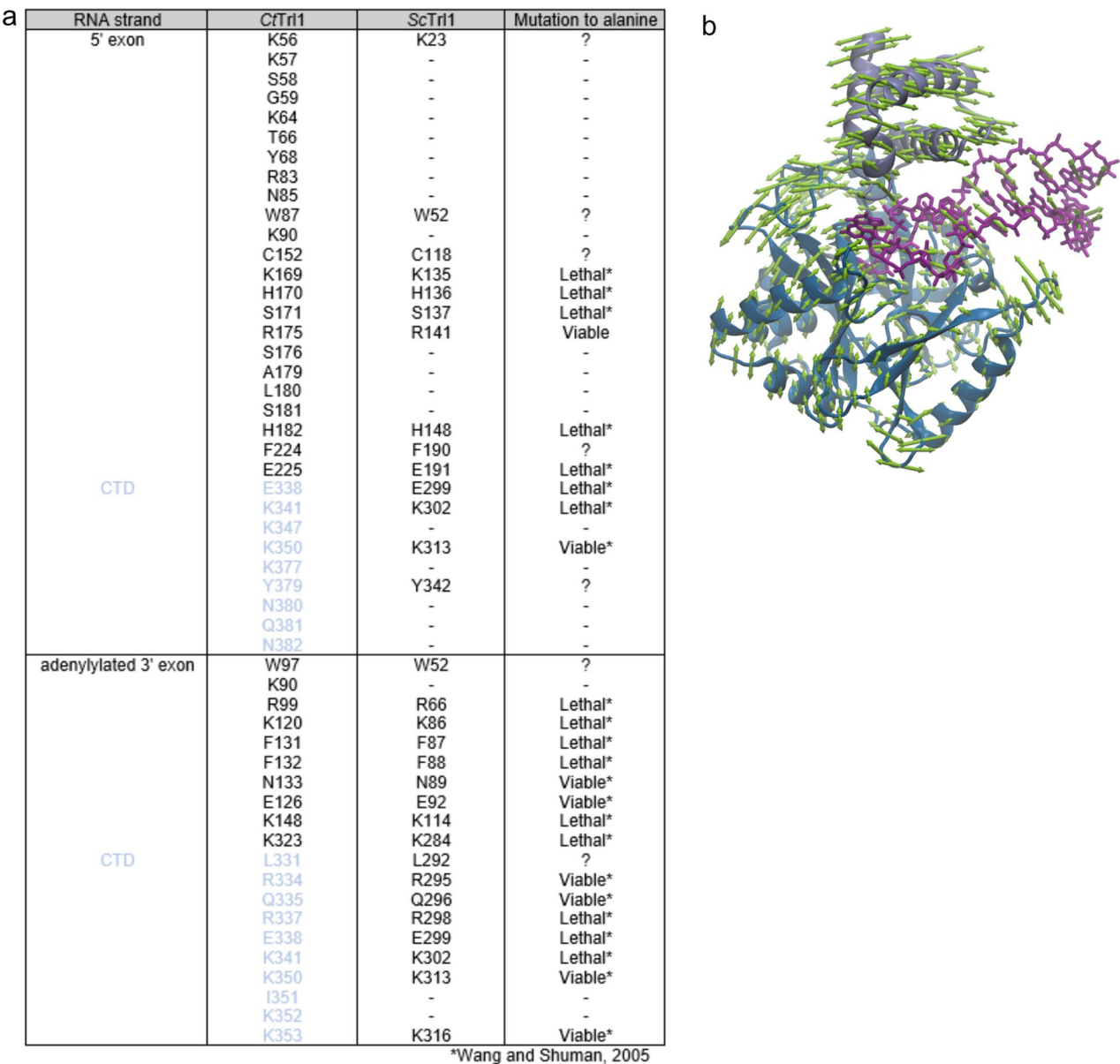
degree of conservation from most variable (green; 1) to most conserved (purple; 9). Additional residue information is given in the legend. The Multiple sequence alignment as well as the *CtTrlI*-LIG structure are shown in the conservation scale colors, respectively. **b**, Electrostatic surface potential map of *CtTrlI*-LIG. Positive potential is represented in blue and negative potential in red.



*Wang and Shuman, 2005

Extended Data Fig. 5 | Impact of RNA-coordinating residues on yeast cell growth. a, Yeast growth assay of the indicated strains to test cell survival of controls and after introducing Trl1 mutations. Dilution series (5-fold) of yeast cells were spotted on the indicated selective solid media. **b,** Overview

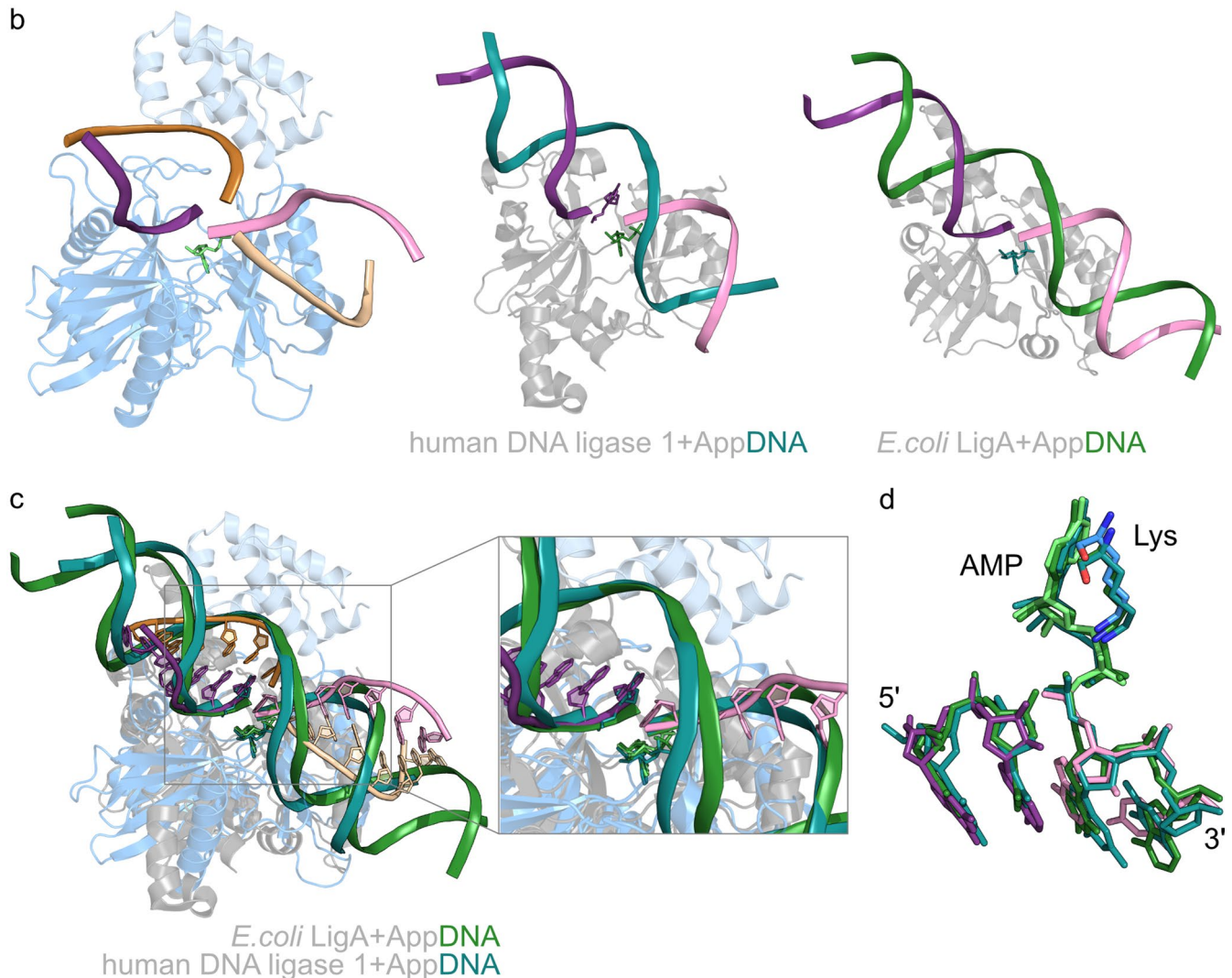
of RNA binding amino acids in *CfTrl1* (based on PDB ID: [8RBJ](#)) as well as the corresponding residues in yeast (*ScTrl1*). The cell viability of the respective alanine variants in *S. cerevisiae* is indicated based on this study or in⁴².



Extended Data Fig. 6 | Modelling of RNA into the active site of CtTrl1-LIG.
a, Overview of RNA binding amino acids in CtTrl1-LIG (based on ASL-RNA modeling in Fig. 4a) as well as the corresponding residues in yeast (ScTrl1). Amino acid residues of the CTD are colored in light blue. The cell viability of the respective

alanine variants in *S. cerevisiae* is indicated based on this study or in⁴². **b**, Mode 1 of Principal Component Analysis illustrates the clamping motion of the protein toward the tRNA. The green arrows indicate the movement directions.

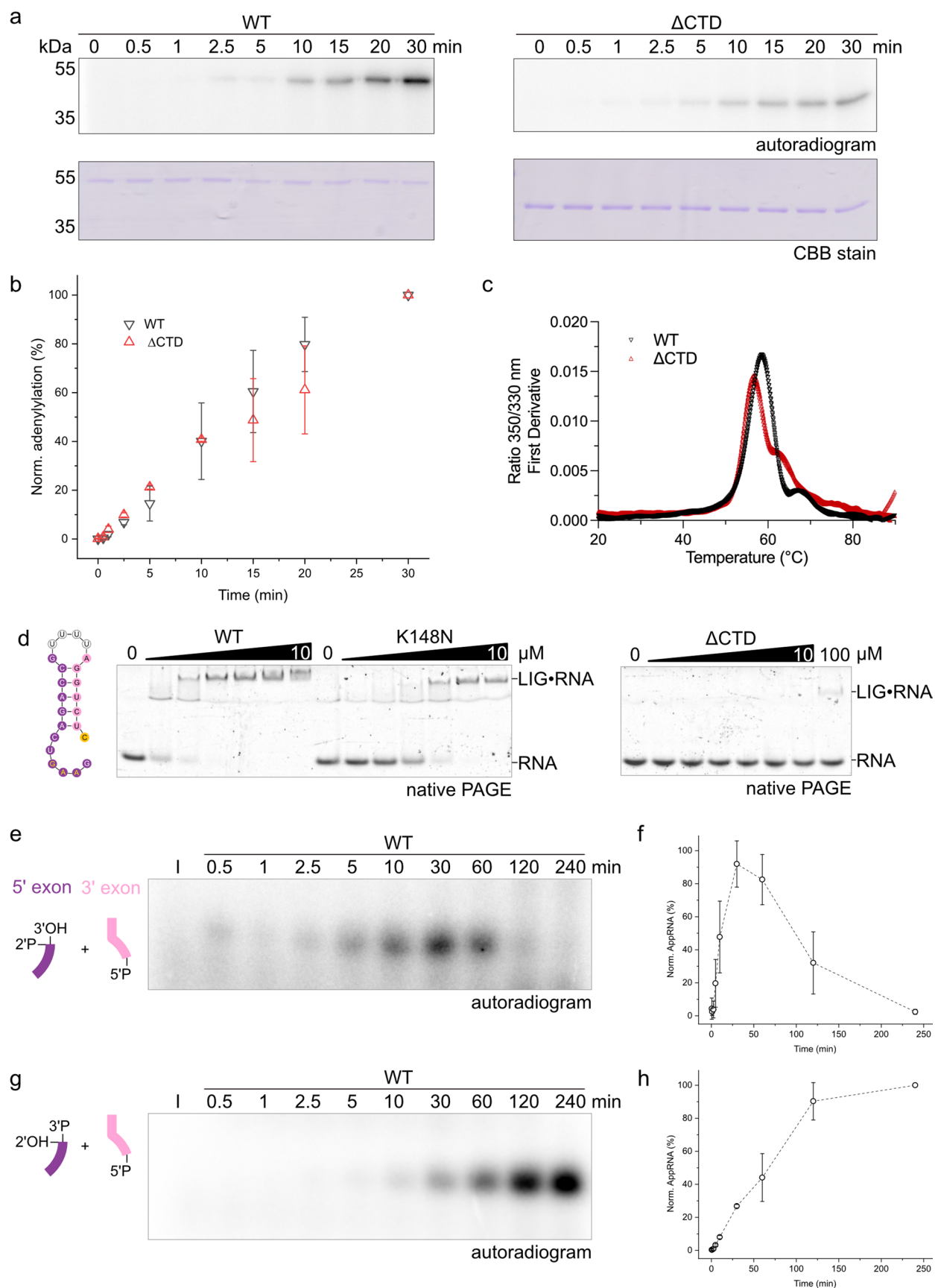
a	CtTrl1-LIG	MDGTAEKQRDSLASRYKASTDYAKDAEGLTAPYVSQDPQETAALVRALDDAAKKGKSGGFSVKTRYAVASSPTGAEDVSWRFNDWDYKPDLPYARGLFTTRTQHGVPETAVRGYDK	120
	T4Rnl2	-----MFKKYSSLEN---HYSNK-----FIEK	19
		*:. . . *	
	CtTrl1-LIG	FFNIDETRDTAWSAIRERTKGPYELTLKENGCIIFISGLEDTLLVCSKHSTGDRS-----DVALSHSSAGEKHLEAQLERIGKTKEELARELRKRNATAVAECCDSFEEHILAYGP	233
	T4Rnl2	LYSLG-LTGGEWVAREKIHGTNFSLIIER-----DKVTCARRTGPILPAEDFFGYEILKNYADSIKAVQDIMETSAVVSIVQVGEFAGFGIQKNVDYCDKDFYVFDIIVTT	125
		:. . . . * * . : . : * : : . : : * : : : * : : . : * : : : * : . : * : * . : .	
	CtTrl1-LIG	DKAGLYLHGINLNIPEFITYPSPLVQKFAEDWGFRTGLIIIDNIDV-----KAFLEE---VAETGAHDG-----RDVEGFVIRCKKSTNPGVGPYHDWFFKYKFEED	329
	T4Rnl2	ESGD-----VTYVDDYMMESFCNTFKFKMAPLLGRGKFEELIKLPNDLDSVVQDYNFTVDHAGLVDANKCVWNAEAKGEVFTAEGYVLK-----PCYPSWLRNGN----	220
		: : * : : * : : * : : : : : : : * : * . : . : * : : : * : : : * : : : *	
	CtTrl1-LIG	YIMYQWRECTKALISGKQP---KIKKHVKITEEYLLYARKRLAADPKLAKLYNQNHGIIKLKRNDFL-----EYKNMKGTDAAAN-LEDDG-----AASV-----	414
	T4Rnl2	---RVAICKNSKFSEKKSDKPIKAKVELSE-----ADNKLVGILACYVTILNRVNVISKIGEIGPKDFGKVMGLTVQDILEETSREGITLTQADNPSLIKELVKMVQDVL	325
		* : * : : * * : * : * : : * : : * : : * : : * : : * : : * : : * : : * : : * : : *	
	CtTrl1-LIG	----- 414	
	T4Rnl2	RPAWIELVS 334	



Extended Data Fig. 7 | Superposition of CtTrl1-LIG•RNA with DNA ligase•AppDNA structures reveals conserved modes of RNA binding.

a, Sequence alignment of CtTrl1-LIG and T4 Rnl2 amino acid sequence. We calculated the sequence identity of CtTrl1-LIG and T4 Rnl2 ligases to 22% with a sequence similarity of 34%. **b**, Side-by-side view of our CtTrl1-LIG•RNA structure with the crystal structures of *E. coli* LigA•AppDNA (PDB ID: 2OWO, grey protein and dark green DNA strands) and human DNA ligase 1•AppDNA (PDB ID: 1X9N, grey protein and dark cyan DNA strands). For better comparison, the 5'

exons were colored in purple and the 3' exons in pink. **c**, Superposition of the structures from (b). The structures were superposed along the β-sheets of the adenylyltransferase domain. The DNA strands are depicted as cartoons. **d**, Zoom into the activated 5' end of the 3' exon and the 3' end of the 5' exon of the nucleic acids in the structures from (c) reveals a conserved coordination principle among adenylyltransferases. The active site lysine residues (K148 of CtTrl1-LIG, K115 of *E. coli* LigA and K568 of the human DNA ligase 1) as well as the first two nucleotides of both strands are depicted as sticks.



Extended Data Fig. 8 | See next page for caption.

Extended Data Fig. 8 | Analysis of the role of the Trl1-LIG C-terminal domain during RNA ligation. **a**, Active site adenylation of WT *CrTrl1*-LIG and the Δ CTD variant was monitored by incorporation of α P³²-labelled ATP. The time course of the adenylation reaction was analyzed by autoradiogram (upper panel). Even protein loading was confirmed by SDS-PAGE and Coomassie Brilliant Blue (CBB) staining (lower panel). **b**, Quantification of the experiment in A performed as technical triplicates ($n = 3$). Data are depicted as mean (circles) including standard deviation (error bars) using Origin 2022b. **c**, Thermostability of WT *CrTrl1*-LIG and the Δ CTD detected with changes in protein intrinsic fluorescence plotted as the first derivative of the ratio between fluorescence at 350 and 330 nm with temperature. Melting peak was detected at 58.5 °C for WT and 56.7 °C for Δ CTD variant. **d**, Electrophoretic mobility shift assay (EMSA) with *CrTrl1*-LIG WT, K148N (both 0–10 μ M) and Δ CTD variants (0–10 and 100 μ M). Increasing amounts of protein were incubated with the ASL-4U-5C RNA (1 μ M) and analyzed by native PAGE using SYBRGold RNA staining. Formation of the LIG•RNA complex

was assessed by band shift. This assay was performed in triplicates ($n = 3$). **e**, Time course of RNA (2'-P and 3'-OH on the 5' exon end and 5'-P on the 3' exon end) adenylation by *CrTrl1*-LIG WT. The autoradiogram shows formation of the adenylylated AMP-ASL-4U intermediate via α P³²-AMP incorporation. **f**, Densitometric quantification of the experiment in (e) performed as triplicates ($n = 3$). Data are depicted as mean (circles) including standard deviation (error bars) using Origin 2022b. The graph reveals a bell-shaped curve for the formation of the AppRNA intermediate over time and its fading through the ligation reaction. **g**, Time course of RNA (2'-OH and 3'-P on the 5' exon end and 5'-P on the 3' exon end) adenylation by *CrTrl1*-LIG WT. The autoradiogram shows formation of the adenylylated AMP-ASL-4U intermediate via α P³²-AMP incorporation. **h**, Densitometric quantification of the experiment in (g) performed as triplicates ($n = 3$). Data are depicted as mean (circles) including standard deviation (error bars) using Origin 2022b. The graph reveals a saturation curve for the formation of the AppRNA intermediate over time.

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Data collection

Diffraction data were collected at the European Synchrotron Radiation Facility (ESRF, Grenoble) using the MxCuBE3 (Macromolecular Xtallography Customized Beamline Environment) Web interface (v3.0.2) and ISPYB (Information System for Protein CrystallographY Beamlines) information management system ISPyB for MXBeta (v5.31.0).

Data analysis

CCP4i software (v8.0.0009), XDS v Feb 05 2021, AIMLESS v1.12.10, PHASER v2.8.3, Phenix v1.19.2_4158, PyMOL v 2.4.1, PISA webserver v1.52, PLIP web tool version 2.2.0, Coot v 0.9.4.1, Affinity Designer v 1.9.2.1035, ConSurf WebServer (2019), Origin 2022b (OriginLab), GraphPad Prism 10.1.1, ImageJ 1.54f, PR.PantaAnalysis software (Version 1.8, NanoTemper Technologies GmbH), LigPlot+ v.2.2, The UNAFold web server (RNA folding form), AMBER22 (2022), eINémo web server (from 25.04.2023), ModeRNA (2009), SimRNA (2015)

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Structure factors and atomic coordinates of the reported X-ray crystallographic structure have been deposited in the Protein Data Bank (PDB) with accession code 8RBJ [https://doi.org/10.2210/pdb8RBJ/pdb]. We have also compared and superimposed our structural models with the following PDB entries: 6N67 [https://doi.org/10.2210/pdb6N67/pdb], 6NOV [https://doi.org/10.2210/pdb6NOV/pdb], 2HVR [https://doi.org/10.2210/pdb2HVR/pdb], 2OWO [https://doi.org/10.2210/pdb2OWO/pdb], 1X9N [https://doi.org/10.2210/pdb1X9N/pdb], 1EHZ [https://doi.org/10.2210/pdb1EHZ/pdb]. The MD simulation trajectory [https://doi.org/10.5281/zenodo.14333445] as well as the top elastic modes of NMA [https://doi.org/10.5281/zenodo.14353453] and the cluster representative for SimRNA simulation [https://doi.org/10.5281/zenodo.14334232] are accessible at Zenodo. The authors declare that the data supporting the findings of this study are available within the paper and its Supplementary Information files. Should any raw data files be needed in another format they are available from the corresponding author upon reasonable request. Source data are provided with this paper.

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